Rheinland-Pfälzische
Technische Universität
Kaiserslautern
Landau

# Multi-Person Human Mesh Recovery with the Anny Body Model

**Master's Thesis**

by

*Muhammad Saad Najib*

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RPTU University Kaiserslautern-Landau
Department of Computer Science
67663 Kaiserslautern
Germany

Examiner: Prof. Dr. Didier Stricker
Muhammad Saif Ullah Khan

## Eigenständigkeitserklärung

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## Use of AI tools

AI assistants (Claude, Anthropic, and Codex, OpenAI) were used under the author's direction for code development, debugging, analysis scripts and assistance with drafting and revising the report. Numerical provenance is recorded in the repository and in Appendix A.

# Abstract

Human mesh recovery estimates a person's 3D body from an image. In this thesis, I study how to make a multi-person model smaller and faster while keeping useful reconstruction accuracy. I use the Multi-HMR framework with Anny, a body model with named shape controls such as age, height and proportions. A smaller ViT-B student learns from a larger ViT-L teacher through knowledge distillation.

The training data combine synthetic AnnyOne images with real images from 3DPW. These datasets use different body models, so I fit Anny bodies to 3DPW's source joints before training. The corrected label set contains 16,694 images and 22,749 labelled person instances. The teacher uses 30 % real images, consistent head-anchor targets and fixed corrections for selected bone rotations. The student learns from these labels and from the teacher's features and body predictions. A trainable projector connects the smaller encoder to prediction heads copied from the teacher.

The student has 101.38 million parameters and requires 603.3 GFLOPs for the recorded one-person input, compared with 318.38 million and 1927.4 GFLOPs for the teacher. In a paired L40S benchmark, a forward pass takes 28.5 ms for the student and 59.7 ms for the teacher. This saving comes with higher reconstruction error. Student PVE is 119.6 mm on the synthetic holdout and 110.1 mm on converted 3DPW; the teacher reaches 73.7 mm and 89.0 mm. The student improves on the released baseline's 126.9 mm 3DPW PVE, although the baseline did not receive the same 3DPW training. Its synthetic result may include images seen in training.

I also compare the models on a walking video and add optional output smoothing and landmark-based alignment. These steps use the trained student without retraining it. The original comparison records a tracked person in all 356 student frames and 354 baseline frames. These counts show coverage, not 3D accuracy. The main limitations are the student's accuracy gap, unreliable child proportions on real photos and remaining placement errors. All 3DPW reconstruction scores use fitted Anny labels, so they cannot be compared directly with published SMPL-space scores.

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# 1. Introduction

## 1.1. Motivation

A photograph shows a person in two dimensions. Human mesh recovery estimates the person's 3D body, including pose, shape and position relative to the camera. When several people appear in the image, the system must also find them and keep their predictions separate. Such models can support motion visualization and applications that need to understand people in a scene.

Multi-HMR [Bar+24] predicts bodies for several people in one forward pass. A Vision Transformer extracts image features, a detection head finds each person's anchor point, and a Human Perception Head predicts body parameters and depth. The original paper uses SMPL-X. NAVER later released an Anny-based version [Bré+26; NAV26b], which I use as a baseline.

Anny has 11 named shape controls and a full rig of 163 bones. Its age and proportion controls can produce both child and adult bodies. However, being able to create a child body does not mean that a network will predict one for a child in a photograph. AnnyOne provides complete labels for synthetic images, but their appearance differs from real photos. Real datasets such as 3DPW provide useful examples, but their body labels use SMPL and need conversion before Anny training.

The other problem is computational cost. A large image encoder can be accurate but slow and expensive to run. I therefore train a smaller student using a larger teacher. The aim is to measure how much model size and inference time can be reduced, and how much accuracy is lost.

## 1.2. Problem Statement and Scope

The task is to predict an Anny body for each detected person from one RGB image and camera intrinsics. Each prediction contains bone rotations, body-shape values and a head-anchor position in camera coordinates. I use three main model names: *Student* is the smaller ViT-B model; *Final* is the larger ViT-L teacher; and *Multi-HMR-Anny* is NAVER's released baseline. Final is the name of a teacher checkpoint, not the name of the deployed student.

The thesis addresses three questions:

1. How can 3DPW's SMPL labels be converted into Anny labels, and which body properties can this conversion recover?

2. How much smaller and faster is the student, and how accurate is it compared with its teacher and the released baseline?

3. How does the student behave on a walking video, and how can output smoothing and alignment improve its use on video?

I evaluate reconstruction on AnnyOne and converted 3DPW labels, and examine the teacher's age and gender predictions on Relative Human. The video application runs the image model on each frame, then optionally filters its outputs across time and adjusts placement using image landmarks. It does not train a video network. Collecting new real-image shape labels and estimating age in years are outside this work.

## 1.3. Contributions

The main contributions are:

1. **Anny labels for real images.** I convert the standard 3DPW splits using joint fitting, virtual torso joints and one shared body shape per person. Independent checks describe the remaining label errors.

2. **A smaller student model.** I combine a ViT-B encoder with a trainable projector, copied prediction heads and feature/output distillation. The student has 101.38 M parameters, compared with 318.38 M in the teacher and 380.94 M in the released baseline.

3. **Evaluation of accuracy and cost.** I compare the three models under the same evaluation conventions and study the teacher's anchor and bone corrections. Body geometry, placement and shape predictions are checked separately because one score cannot describe all three.

4. **A video comparison tool.** I save the input and predicted meshes side by side, with optional student smoothing and landmark alignment. The tool also records missing predictions, changes between frames and inference timings.

The student is 3.14 times smaller than the teacher and 2.10 times faster in the paired L40S benchmark, but the teacher remains more accurate on both reconstruction sets. The student's lower 3DPW error than the released baseline includes the benefit of additional real-image training; it cannot be attributed to the smaller architecture alone.

## 1.4. Thesis Structure

Chapter 2 introduces the body models and related work. Chapter 3 explains the datasets and label conversion. Chapter 4 describes the model, training losses and video processing. Chapter 5 reports the results, and Chapter 6 discusses what they show and where they are limited. Chapter 7 concludes with possible next steps. The appendices contain reproduction details and the supporting diagnostic experiments.

# 2. Background and Related Work

This chapter explains the body models and earlier methods used in this work. I focus on the ideas needed to understand the label conversion, student training and video application.

## 2.1. Parametric body models: the SMPL family and Anny

A parametric body model is a differentiable function $M(\boldsymbol{\theta}, \boldsymbol{\beta}) \to \mathbf{V} \in \mathbb{R}^{N\times 3}$ that maps a pose $\boldsymbol{\theta}$ and a shape $\boldsymbol{\beta}$ to the $N$ vertices of a mesh. The shape deforms a template mesh in its rest pose. The pose moves a skeleton of bones, and the skin follows by *linear blend skinning* (LBS):

$$\mathbf{v}'_i \;=\; \sum_k w_{ik}\, \mathbf{G}_k(\boldsymbol{\theta}, \boldsymbol{\beta})\, \mathbf{v}_i(\boldsymbol{\beta}), \tag{2.1}$$

where $\mathbf{v}_i(\boldsymbol{\beta})$ is the shaped rest vertex, $w_{ik}$ the skinning weight of bone $k$ and $\mathbf{G}_k$ the bone's world transform, found by chaining the rotations of all bones from the root down to $k$. Because the transforms chain, a wrong rotation high in the tree moves everything below it. This matters in Appendix F, where two neck bones turned by half a turn leave the head pointing the right way but twist the skin between them.

**The SMPL family.** SMPL [Lop+15] has 6,890 vertices and 24 joints. Its shape space is a set of principal components learned from registered scans of adults; most methods use the first 10 coefficients. These are statistical directions, not named properties: no coefficient means "older" or "more muscular". Adult scan statistics limit the representation of child proportions. SMIL [Hes+18] is a separate infant model, and AGORA [Pat+21] and BEV [Sun+22] blend the adult template towards an infant one with one extra coefficient. SMPL-H [RTB17] adds MANO hands with 15 joints each, and SMPL-X [Pav+19] adds a face, for 55 joints on 10,475 vertices. SMPL is provided under research licence terms; the terms for assets, datasets and trained weights must be checked separately. The original Multi-HMR paper predicts SMPL-X [Bar+24]. This thesis uses the later Anny-compatible release and its body-model interface.

**Anny.** Anny [Bré+26] is a body model by NAVER LABS Europe built from the open MakeHuman project [Mak] instead of from scans. Its code is under the Apache 2.0 licence and its assets under CC0 [NAV26a]. The implementation



**Figure 2.1.:** *Anny at the mean AnnyOne shape in rest pose, with only the age value changed from 0 to 1. The skeletal height (head joint to mean ankle) grows from 0.53 m to 1.54 m. Body model only, no network; one camera for all panels, so the relative size is visible. The approximate years printed in the panel titles come from one of three age scales that disagree (Section 2.1) and are not used in this thesis.*

uses the full rig of 163 bones, compatible with the AnnyOne labels. The body-model package and topology must match the checkpoint and annotation configuration; the parameter count of a default rig in another release is not a substitute for that configuration. Pose is one rigid transform per bone, applied by Equation (2.1). Of the bones, 88 are helper bones (toes, breasts, face micro-bones) whose rotation an image cannot reveal; I keep them at rest, which leaves 75 predicted bones. The neck is modelled by three stacked bones, `neck01` to `neck03`, followed by `head`.

Shape is set by 11 *phenotype* values between 0 and 1: `gender`, `age`, `muscle`, `weight`, `height`, `proportions`, `cupsize`, `firmness`, and three blend weights `african`, `asian` and `caucasian`. Each value drives artist-made blend shapes, vertex offsets of the rest mesh. Because the values have names, a predicted body can be read as "older" or "heavier". Gender 0 is the male end and 1 the female end (rest stature 1.753 m and 1.632 m). Age blends five anchor bodies placed at $\mathrm{linspace}(-\frac{1}{3}, 1, 5)$: age 0 is the baby, 0.333 the child, 0.667 the young adult and 1 the old adult; the newborn would need an age below 0, which a network output squashed into $(0, 1)$ never gives.

Figure 2.1 shows what age does. At the mean AnnyOne shape, moving age from 0 to 1 raises the skeletal height from 0.53 m to 1.54 m (1.37 m at 0.5); most of the change happens below about 0.5. A small height value does not make a child: at age 0.6 the smallest reachable body is 1.22 m tall and keeps adult proportions. What marks a child is narrow shoulders relative to the head, a shoulder-to-head width ratio of 2.24 at age 0.1 against 3.00 at 0.6. Only a low age produces it, which is why Appendix F treats children as an age problem. I never convert ages to years, because three scales are in use and disagree: the value read as 11 years is 0.415 in Anny's calibration, 0.333 in the code I inherited and 0.19 on the MakeHuman slider. I call a person *child-shaped* if the age value is below about 0.45.

**AnnyOne.** AnnyOne is NAVER's synthetic dataset for Anny: rendered scenes of several people with exact Anny labels (all bone transforms, all phenotypes and the camera). Its authors describe more than 780,000 images [Bra+26]; the release copy I train on holds 579,960 images. It is the only source of complete Anny labels, shape included, that I have (Section 3.1).

## 2.2. Image Features and Attention

A Vision Transformer divides an image into patches and represents each patch by a token [Dos+21]. Attention combines token information through queries $\mathbf{Q}$, keys $\mathbf{K}_a$ and values $\mathbf{V}_a$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}_a, \mathbf{V}_a) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_a^\top}{\sqrt{d_k}}\right)\mathbf{V}_a. \tag{2.2}$$

The subscript distinguishes attention keys from camera intrinsics $\mathbf{K}$. Self-attention relates image patches to each other; cross-attention allows a person query to read information from image tokens. Multi-HMR uses both image features and person queries, enabling one shared encoding of a multi-person image. DINOv2 provides pretrained visual features for the ViT-L teacher and ViT-B student [ODM+23]. A smaller encoder reduces cost but changes feature width, motivating the projector and transferred-head design in Chapter 4.

## 2.3. Single-person and multi-person human mesh recovery

**Regression and fitting.** HMR [Kan+18] regressed body-model parameters for one cropped person with a convolutional encoder, an iterative regressor and an adversarial pose prior. HMR 2.0 [Goe+23] replaced the encoder by a Vision Transformer (ViT) [Dos+21] and showed that a large ViT trained on a broad data mixture is a strong and simple recipe, which Multi-HMR shares. My image model also predicts the parameters directly. The optional video alignment step is a separate optimization of placement after this prediction. SMPLify [Bog+16] goes the other way and fits SMPL to 2D keypoints by optimisation; SPIN [Kol+19] puts such a fit inside the training loop. My 3DPW converter (Chapter 3) is an optimisation of the SMPLify kind, but in 3D, between two body models, and run once offline.

**Single-shot multi-person: ROMP and Multi-HMR.** Two-stage methods run a detector and then a single-person network on each crop, so the cost grows with the number of people. ROMP [Sun+21] does both in one pass: it predicts a heatmap of body centres and reads body parameters at the detected centres. Multi-HMR [Bar+24], the architecture of this thesis, keeps this idea. A DINOv2 [ODM+23] ViT turns the image into one token per 14×14-pixel patch, a 48×48 grid at 672 pixels. Each token scores whether a person's primary keypoint, the head joint in the released code [NAV26b], lies in it. High-scoring tokens become the queries of the *Human Perception Head* (HPH), which attends to all image tokens and regresses the body parameters and the person's distance from the camera. That distance belongs to the head joint, and this anchor is at the centre of the depth defect of Appendix F. Section 4.2.2 gives the details. The released SMPL-X checkpoints reach a PVE of 94.1 mm on the 3DPW test set with ViT-L in 74 ms and 94.0 mm with ViT-B in 43 ms on a

V100 [NAV26b], which is why my smaller model is a ViT-B student. These are SMPL-X errors and cannot be compared with the Anny-space errors of this thesis.

**BEV and Relative Human.** BEV [Sun+22] extends ROMP with explicit depth reasoning and an age-aware body. Its authors collected Relative Human, real photos in which every person carries an age group (adult, teenager, kid, baby), a depth layer and keypoints. I use its real-photo age-group and gender annotations for evaluation only (Section 3.7). It provides no reference 3D bodies for this evaluation.

**Multi-HMR-Anny and Anny-Fit.** In February 2026 NAVER released a Multi-HMR checkpoint that predicts Anny, trained on BEDLAM, AnnyOne and further real images [NAV26b]. A multi-person HMR network for Anny therefore already exists, and this thesis does not claim to be the first. This checkpoint, Multi-HMR-Anny, is where my backbone weights come from and the baseline I compare against (Section 5.2). On the same backbone its decoder has 8 cross-attention blocks and 76.57 M parameters, against 2 blocks and 14.01 M in mine. It was trained on AnnyOne, whose release includes the images I hold out. Its exact image exposure is not verified here, so I treat its holdout score as potentially in-sample. Anny-Fit [Bra+26], by the same group, refines such predictions per image by optimization and adapts them to children at test time. I did not evaluate it. It would be a useful comparison for future work on children; the video's placement-only alignment does not perform this shape fitting.

## 2.4. Synthetic training data

AGORA [Pat+21] renders scanned people with SMPL-X fits into realistic scenes, and BEDLAM [Bla+23] showed that synthetic images alone can train competitive networks. AnnyOne does the same for Anny. Synthetic labels are exact and complete, but leave open how well a network transfers to real photos. In this thesis my shape predictions follow the labels on the synthetic holdout but collapse towards the training average on real photos (Appendix F). This is why Final mixes 30 % real images into every batch, and why those images need Anny labels.

## 2.5. Real-data labels for a new body model

3DPW [Mar+18] is the standard real-world test set for mesh recovery: 60 video sequences with SMPL ground truth for one or two actors each. SMPL and Anny share neither skeleton nor shape space, so the labels must be re-expressed in Anny by fitting. The SMPL-X code transfers fits between SMPL-family models through a vertex correspondence [Pav+19], and the Anny package offers an inverter that fits Anny to a target mesh [NAV26a]; none turns a 3DPW label

into an Anny label directly, so I wrote my own converter (Section 3.3). A fit to joint positions recovers only what joint positions determine; what the converted labels can and cannot test is listed in Section 3.6. SHAPY [Cho+22] supervises shape on real images with measurements and word attributes instead; Anny's named phenotypes would suit such labels, but I collected none.

## 2.6. Knowledge distillation

Knowledge distillation [HVD15] trains a small *student* to reproduce a large, trained *teacher*, originally by matching its class probabilities softened by a temperature. FitNets [Rom+15] let the student also match an intermediate representation of the teacher through a small learned regressor. DINOv2 [ODM+23] distilled its smaller ViTs from its largest one, so a DINOv2 ViT-B is a good starting point for a student. My Student combines both forms: a projector maps its 768-wide ViT-B tokens to the teacher's 1024-wide tokens, which lets the teacher's heads be copied unchanged (Section 4.8). It has 101.38 M parameters against 318.38 M, and I test it against the error-retention criterion defined in Section 4.8.1.

## 2.7. Temporal models for video

The student predicts each image independently. To reduce changes between frames, I add optional filtering of its outputs. Learned video methods go further by training on sequences. Several refine per-frame features of an image network: HMMR [Kan+19] with a temporal convolution, VIBE [KAB20] with a recurrent network and a motion discriminator tied to SMPL, TCMR [Cho+21] with branches that predict the current frame from its neighbours to reduce jitter, and MPS-Net [Wei+22] and GLoT [She+23] with attention over frames. SmoothNet [Zen+22] smooths the output poses of any estimator without image features. PHALP [Raj+22] and 4DHumans [Goe+23] track people through video, the first step of any multi-person video method, and WHAM [Shi+24] places the motion in world coordinates.

My video application keeps the raw predictions as a reference and adds tracking, exponential smoothing and optional landmark alignment (Section 4.10). The filter uses previous outputs without learning new weights. It is simpler than the learned temporal methods above, but can introduce delay.

## 2.8. Positioning the Present Work

The method follows the single-shot detection and regression design of Multi-HMR. Its compactness comes from a smaller image encoder with a projector and transferred heads, rather than a new attention operator. The projector serves two roles: it aligns feature dimensions for distillation and preserves the representation width expected by the copied prediction heads. This differs

| Approach | Relevant idea | Relation to this thesis |
|---|---|---|
| Multi-HMR and its released Anny model [Bar+24; NAV26b] | single-shot multi-person detection and body regression | baseline framework; source checkpoints; external comparison |
| FitNets and knowledge distillation [Rom+15; HVD15] | representation and output transfer from a teacher | motivate projected feature matching and parameter distillation |
| BEV [Sun+22] | age-aware reconstruction and real-photo age-group annotations | motivates evaluating child proportions; not an implemented student component |
| VIBE, TCMR and SmoothNet [KAB20; Cho+21; Zen+22] | temporal features or output refinement | possible learned alternatives to the implemented output filters |

**Table 2.1.:** *Relevant prior work and its role. The method combines an existing single-shot framework with compact teacher–student training and converted Anny supervision.*

from a feature adapter used only during training; the deployed student keeps the projector.

Real-image supervision is another distinct part of the system. Using an existing body model does not supply annotations in its parameter space. Joint-based conversion makes 3DPW useful for Anny training, but cannot recover shape variables that the source joints do not constrain. This limitation motivates separate reconstruction and phenotype evaluations.

# 3. Datasets and Anny Label Generation

I use AnnyOne for complete synthetic labels, 3DPW for real images, and Relative Human to check shape predictions on real photos. Table 3.1 lists the splits and their roles. The main preparation step is converting 3DPW's SMPL annotations to Anny. The quality and limits of these fits affect how the real-image results should be read. I exclude the test images from training in this thesis; the released baseline's earlier exposure to AnnyOne is a separate limitation.

| split | size (images / people) | used for | chapters |
|---|---|---|---|
| AnnyOne training | 579,860 images | training of thesis models; 70% of each mixed batch | 4, 5 |
| AnnyOne holdout | 100 images | synthetic evaluation (V1); excluded from thesis training | F, 5 |
| diagnostic manifest | 40 + 15 entries | per-person diagnostics; excluded from every fine-tune | F |
| 3DPW-Anny train, v23 | 5,882 / 7,584 | 30% of each mixed batch | 4, 5 |
| 3DPW-Anny val., v23 | 2,622 / 3,311 | 300 images scored after each epoch; monitoring only | 5 |
| 3DPW-Anny test, v23 | 8,190 / 11,854 | reported 3DPW numbers; never trained on | 5 |
| 3DPW-Anny old labels | 16,631 / 20,028 | historical experiments only; not used by Student | E |
| Relative Human val. | 635 / 2,130 | children and gender on real photos; evaluation only | F, 5 |
| demo images | 12 images | by-eye comparison; no metric | F, 5 |

**Table 3.1.:** *Every split of the thesis, its size and its use. 3DPW-Anny sizes are images / labelled people after the drop rules (v23 labels, Section G.2); numbers on 3DPW are always against these converted labels, never against the original SMPL meshes. The reported teacher and 30-epoch student use the last checkpoints of their specified schedules; validation monitors training.*

## 3.1. AnnyOne

AnnyOne [Bré+26] is the synthetic dataset released with Anny. It holds 579,960 rendered images of 1280×1280 pixels, with 2.73 people per image on average. For every visible person the label gives the exact Anny pose (163 bone transforms), the 11 phenotypes and the camera intrinsics. The network

sees each image resized to 672 pixels. The field of view changes from image to image (51° to 110° on a sample of five images), so training and evaluation use each image's own intrinsics.

**Split.** Indices 0 to 579,859 are the training set. The last 100 images, indices 579,860 to 579,959, are the holdout, my own choice and not an official split; no model trained in this thesis has seen them. NAVER's model was trained on the AnnyOne release, so that model may have seen these images before. Its exact exposure is not verified. The 3DPW test script also scores the last 100 AnnyOne images, but in setting V0; its "AnnyOne test-100" value is therefore not the holdout number, which is always given in setting V1.

**Few children.** In Anny the child shape appears below an age of about 0.4 to 0.5 (Section 2.1). Only 2.53% of AnnyOne's people are below an age-parameter value of 0.3 (1.06% below 0.2, 4.45% below 0.4); the median age is 0.601, and 41% of all people lie between 0.5 and 0.6. In a training batch, 2.1% to 2.3% of the supervised people are younger than 0.3. The holdout has nobody younger than 0.513 and tests adults only, which is why I added Relative Human (Section 3.7).

## 3.2. 3DPW

3DPW [Mar+18] is a real video dataset of 60 sequences filmed with a hand-held phone camera at 1080×1920 pixels (portrait) and 30 frames per second. Inertial sensors on the actors give SMPL ground truth. The standard release has 24 training, 12 validation and 24 test sequences, with one or two labelled actors each: 87 person-sequences (62 men, 25 women). Bystanders have no labels. I keep the splits as 3DPW defines them. One validation sequence, `outdoors_crosscountry_00`, yields no labels (Section G.2), which leaves 11 usable validation sequences.

## 3.3. Converting SMPL to Anny

3DPW describes bodies with SMPL parameters, which cannot be copied directly into Anny. I instead adjust an Anny body until its joints are close to 3DPW's source joints. The converter does this for each labelled person and selected frame. A label builder then moves the fitted bodies into camera coordinates and puts all labelled people from the same image into one annotation file.

**Objective.** I use gradient-based optimization to reduce the distance between corresponding joints. The variables are the permitted bone rotations $\mathbf{r}$ (represented in 6D), the free shape logits $\boldsymbol{\ell}$ and a translation $\mathbf{t}$. The target is the set

of 24 source SMPL joints $\hat{\mathbf{J}}^S$:

$$\min_{\mathbf{r},\boldsymbol{\ell},\mathbf{t}} \frac{1}{24}\sum_{k=1}^{24}\left\|\mathbf{p}_{\pi(k)}(\mathbf{r},\sigma(\boldsymbol{\ell}))+\mathbf{t}-\hat{\mathbf{J}}^S_k\right\|_2^2 + \lambda_r \sum_{b\neq\text{root}} \|\mathbf{R}_b-\mathbf{I}\|_F^2 + \lambda_\phi \sum_{j\in\text{free}} \left(\sigma(\ell_j)-\tfrac{1}{2}\right)^2, \tag{3.1}$$

where $\pi$ is the joint correspondence and $\mathbf{p}_b$ the point attached to bone $b$: its head, or a virtual joint (Section 3.4). The first regulariser ($\lambda_r = 10^{-4}$) pulls every non-root rotation towards identity, because a twist about a bone's own axis moves no joint but corkscrews the mesh. The second ($\lambda_\phi = 10^{-3}$) pulls the free phenotypes towards 0.5, so that one remaining joint gap cannot drag a phenotype to an extreme. Adam (learning rate 0.05) fits the root, then the bone rotations, then the free phenotypes; shape comes last because it is the weakest signal. Each frame starts from the previous frame's solution (150 / 400 / 250 iterations per stage on a person's first frame, 20 / 80 / 60 afterwards). The targets are 3DPW's own per-frame `jointPositions`; the neutral SMPL model would move them by 37 mm on average, because 3DPW's shape coefficients belong to the gendered models.

**Bones and shape.** Only the root and 18 of the 23 mapped bones may rotate; 144 of 163 bones stay at rest, among them `neck02`, `neck03`, `head` and every finger bone. Joints cannot determine most phenotypes: weight, muscle, cup size, firmness and the three ethnicity values move no joint, and adult age barely does. The converter therefore reads gender from 3DPW, fixes age at 0.55 and the seven others at 0.5, and fits only height and proportions. Since version 2.3 it writes men as 0 and women as 1, Anny's convention (`-gender_convention anny`). Height and proportions are estimated by adjusting Anny's controls to bring its joints closer to the source joints; they are not copied from SMPL's shape coefficients. After the first pass, I take the median height and proportion values across the sequence. I use this one shape for the person and fit each frame's pose again (100 iterations). This prevents the same actor from receiving a different body shape in every frame.

**Frames, camera and label files.** I fit every third valid frame (10 frames per second). Since version 2.3 one grid holds for every person of a sequence (frame numbers divisible by 3, `-frame_grid shared`), so both actors are fitted on the same images. The builder writes one file per image in the AnnyOne format and drops a person when the fit error exceeds 80 mm, less than 30% of the mesh is inside the image, or the box is narrower or shorter than 40 px. A dataset class pads the portrait images to a square and shifts the principal point, so the training and evaluation code run on these labels unchanged.

## 3.4. Geometric Correspondence and Virtual Joints

A joint called "hip" does not necessarily sit at the same place in SMPL and Anny. Matching joint names alone can therefore make the fitter bend or stretch the body to correct a difference in joint definitions. The final procedure aligns the rest skeletons using the hip midpoint and body axes, chooses hip bones by geometric proximity, and selects the spine correspondence by normalized height along the torso. Appendix B gives the correspondence construction.

For a torso target $k$ attached to Anny bone $b = \pi(k)$, the fitter uses a virtual joint

$$\mathbf{p}_b(\mathbf{R}, \boldsymbol{\phi}) = \mathbf{h}_b(\mathbf{R}, \boldsymbol{\phi}) + \mathbf{Q}_b(\mathbf{R}, \boldsymbol{\phi})\boldsymbol{\delta}_k, \tag{3.2}$$

where $\mathbf{h}_b$ is the posed bone head, $\mathbf{Q}_b$ its orientation, and $\boldsymbol{\delta}_k$ a constant offset expressed in the bone's rest frame. The offset is measured from the aligned rest skeletons. Rotation moves the virtual point with its bone, so the objective compares equivalent locations without altering Anny's skeleton or mesh. In simple terms, the virtual joint is a marker attached to a bone: it moves with the bone and provides a more comparable fitting point. Limb targets without offsets use the bone head itself. The two body models still have different proportions.

## 3.5. Label Construction and Validation

The final labels use the corrected gender convention (0 for the male end and 1 for the female end in Anny), a shared frame grid for every actor in a sequence, and checks that the corresponding image exists. A shared grid is important for multi-person detection: all labelled actors must appear in the same image annotation. Each actor receives one fitted shape shared across the sequence, while pose varies by frame.

An independent check reads the completed annotations rather than the optimizer's internal loss. It compares body-joint positions, projected keypoints, actor completeness and shape convention against the source data. The final check gives a median body-joint discrepancy of 14.1 mm and a diagnostic status of WARN. The remaining warnings concern limb proportions and joint-definition differences, including the hip, rather than the corrected label-integrity defects. The sample does not prove that every annotation is error-free.

The main experiments use only the corrected v23 labels. Historical converter comparisons and the tests that identified annotation defects are retained in Appendix G. Residual fitting error is a label quality indicator and must not be subtracted from a network's PVE as if it were a universal error floor.

## 3.6. What the 3DPW labels can and cannot test

The v23 labels are fitted estimates. They are the training and evaluation targets used here, but they do not provide an independent measurement of every body property. Four limits matter.

| | old labels | rebuilt | v23 labels |
|---|---|---|---|
| converter fits | v2.2 | v2.2 | v2.3 |
| image files of 8 sequences | missing | re-extracted | re-extracted |
| train, images / people | 4,451 / 5,183 | 6,437 / 7,589 | 5,882 / 7,584 |
| validation | 2,458 / 2,995 | 2,777 / 3,314 | 2,622 / 3,311 |
| test | 9,722 / 11,850 | 9,722 / 11,850 | 8,190 / 11,854 |
| total | 16,631 / 20,028 | 18,936 / 22,753 | 16,694 / 22,749 |
| gender reversed | yes | yes | no |
| actor missing in two-person frames | yes | yes | no |
| labels missing (no image file) | yes | no | no |
| used for training | history models only | never | Final, Control, Student |

**Table 3.2.:** *The three 3DPW-Anny label sets. All are built at every third frame with the same drop rules. The v23 labels have fewer images than the rebuilt set but the same people, because both actors of a two-person sequence now share each frame.* `outdoors_crosscountry_00` *yields no labels in any set (Section G.2). Numbers measured on the old labels are retired and never appear next to numbers on the v23 labels.*

| check | old labels | v23 labels |
|---|---|---|
| overall grade | FAIL | WARN |
| body joints vs 3DPW, median / p95 (mm) | 18.7 / 24.5 | 14.1 / 21.1 |
| surface, Anny→SMPL / SMPL→Anny (mm) | 18.9 / 18.3 | 18.9 / 16.1 |
| 2D error over raw 3DPW (px) | +1.1 | +0.4 |
| gender labels | reversed | men 0, women 1 |
| refit with gender flipped | better for 72 of 73 men | worse: men 13.9 → 20.0 mm |
| two-person images, an actor unlabelled | 407 of 960 | 0 of 1,080 |
| fitted frames; median frame error (mm) | 22,915; 18.2 | 22,912; 14.0 |
| head turn vs chest, labels / SMPL (°) | 9.5 / 20.2 | 19.6 / 19.3 |
| *post-check of the re-fit, all fitted frames, 87 people* | | |
| mean joint error, men / women (mm) | 19.0 / 15.0 | 14.7 / 15.2 |
| two-person frames, a valid actor unfitted | 4,457 of 8,274 | 0 of 6,055 |
| *still WARN on the v23 labels* | | |
| limb ratio Anny / SMPL, forearm / thigh | | 0.948 / 1.064 |
| people with a joint more than 100 mm off | | 3.2% |

**Table 3.3.:** *Independent conversion check of the 3DPW-Anny labels before (old labels, v2.2 fits) and after the re-fit (v23 labels, v2.3 fits). The check samples images from every sequence and compares the labels with 3DPW's own joints, surface and 2D keypoints. The middle block is the re-fit's own post-check over all fitted frames; old and new fits fitted different frames, so it is not a paired comparison.*

**Eight constant phenotypes.** The labels provide person-specific values for three phenotypes: gender (from metadata), height and proportions (fitted; medians 0.47 and 0.51). The other 8 are the same for every person: age 0.55 and seven values at 0.5. A model trained on 3DPW is pulled towards these constants for every real person, by the shape loss and the mesh terms; 3DPW people are about 16% of the supervised people in a 70/30 batch (my estimate). The 3DPW error is also measured against these neutral-body meshes, so it likely rewards a neutral prediction; I did not test this. Section 4.7 describes the shape mask that removes the shape loss on the constants (Final-masked). At age 0.55 every label is a full adult (Figure 2.1); 3DPW has no children, so it can neither teach nor test them.

**Head and hands at rest.** `neck02`, `neck03` and `head` stay at identity in every label, so the head turns only through `neck01`, and the labels cannot test the half-turned neck bones of Appendix F. 3DPW's SMPL ground truth has no finger joints, so every label has rest-pose hands; the training masks their rotation terms for 3DPW people (Section 4.7).

**Residual annotation discrepancy.** The checked labels differ from 3DPW's source body joints by 14.1 mm at the median and 21.1 mm at the 95th percentile. This describes the conversion, not a universal lower bound on network error: a network is compared with the fitted Anny targets, rather than directly with those source joints.

**Not comparable with published numbers.** I never evaluated against the original SMPL meshes, and our MPJPE averages all 163 Anny joints (Section 4.9), so I write "lowest error on our 3DPW-Anny labels". Bystanders are unlabelled, so precision on 3DPW is a lower bound.

## 3.7. Relative Human

Neither the holdout nor 3DPW contains children. Relative Human, introduced with BEV [Sun+22], fills this gap. It holds 8,741 real photos, many crowded and with children. Each person has a box, an age group (adult, teen, kid, baby) and a gender coded 0 for male, as in Anny; there are no 3D labels. The copy used here has no supplied licence file; redistribution terms require separate checking. The experiment uses only the validation split for evaluation: 635 images with 2,130 labelled people (1,680 adults, 101 teens, 209 kids, 117 babies, 23 of unknown age).

The model runs once on each whole photo with a field of view of 55°, and detections are paired with labelled boxes as in Section 4.9. A predicted age below 0.45 counts as child-shaped, a predicted gender below 0.5 as male. I report recall and mean predicted age per age group, the child-against-adult AUC of the predicted age (kids and babies against adults) and the gender accuracy. Every number covers detected people only, so recall always stands next to it.

## 3.8. The diagnostic manifest and the demo images

The checks of Appendix F need the same people every time. A seeded script picks them from the AnnyOne labels alone and stores 55 fixed (image, person) entries: 40 from the training range (adults, the youngest children, clearly visible hands, the hardest head turns) and 15 from the holdout. Base saw the training entries; every later fine-tune gives their images a sampling weight of zero, so they remain the same in-sample reference for all models. The manifest children therefore show whether a model can fit a child it has seen, not whether it generalises. For by-eye comparisons I use 12 real images: 4 photos and 8 3DPW test frames, one middle frame from each of 8 test sequences. No model trained on them and no metric is computed on them.

# 4. Methodology

This chapter explains how I trained the ViT-B Student and used it for image and video inference. The larger ViT-L teacher, called Final in the experiment records, provides the copied prediction heads and teacher targets. The supporting diagnostic experiments are in Appendix F.

## 4.1. System Overview

The pipeline has three stages. First, the conversion procedure of Chapter 3 supplies Anny targets for real 3DPW training images. Second, a larger teacher is trained with synthetic and real supervision. Third, the student learns from both the labels and the frozen teacher. Deployment uses the student alone.

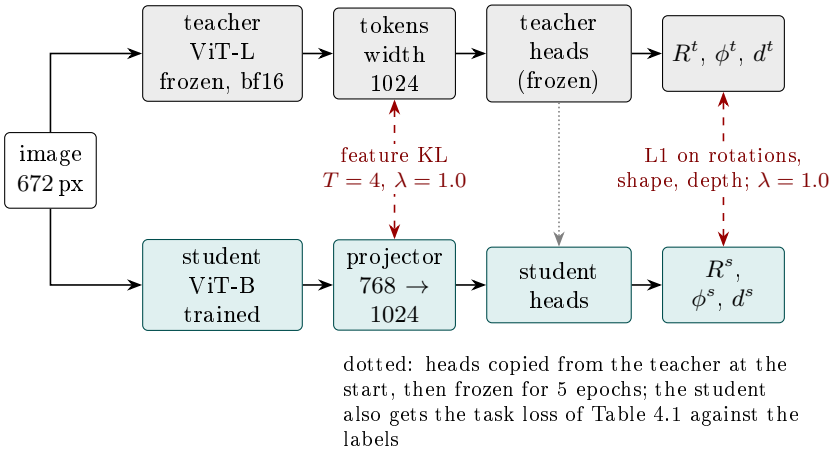



**Figure 4.1.:** *Teacher–student system. Both branches receive the same image. The frozen teacher provides features and per-person outputs. The student's ViT-B features pass through a 768-to-1024 projector and heads initialized from the teacher. The student receives supervised, feature-distillation and output-distillation losses. Only the student branch is needed at inference; its projector remains because the prediction heads use its output.*

For image $I$, student tokens are $\mathbf{Z}^s = f_B(I) \in \mathbb{R}^{N_p \times C_s}$, with $C_s = 768$. The trainable projection is

$$\widetilde{\mathbf{Z}}^s = \mathbf{Z}^s \mathbf{W}^\top + \mathbf{b}, \qquad \mathbf{W} \in \mathbb{R}^{C_t \times C_s}, \quad C_t = 1024. \tag{4.1}$$

For each image patch, the student produces 768 feature values. The projector learns 1024 weighted combinations of those values, plus a bias. It changes their representation so that the teacher's copied heads can use them. It does

not add new image information. Both detection and body prediction use the projected features, so the projector remains part of the student at inference. Most of the saving comes from replacing the larger encoder.

## 4.2. Multi-HMR for Anny: task and architecture

### 4.2.1. The task

The input is one RGB image with camera intrinsics $\mathbf{K}$ (from an assumed field of view when they are unknown). It shows an unknown number of people. For each person $i$ the network predicts the bone rotations $\mathbf{R}_i \in SO(3)^{163}$ of Anny's bones, the phenotypes $\boldsymbol{\phi}_i \in [0,1]^{11}$, and the image position $(u_i, v_i)$ and depth $z_i$ of one anchor joint $a$, the head. The Anny body model $M$ [Bré+26] turns $(\mathbf{R}_i, \boldsymbol{\phi}_i)$ into vertices and joints, and the mesh is moved so that its anchor joint sits at the back-projected anchor pixel:

$$\mathbf{V}_i = M_V(\mathbf{R}_i, \boldsymbol{\phi}_i) - M_{J,a}(\mathbf{R}_i, \boldsymbol{\phi}_i) + \mathbf{t}_i, \qquad \mathbf{t}_i = z_i\, \mathbf{K}^{-1}\,(u_i, v_i, 1)^\top. \tag{4.2}$$

The decoder first builds the body, subtracts its head-joint position and moves the whole mesh to $\mathbf{t}_i$. Its head then sits at the predicted 3D anchor. This changes placement without changing the body's pose or proportions.

### 4.2.2. Architecture

The teacher follows Multi-HMR [Bar+24] with Anny-compatible outputs (Figure 4.2). The student retains the detection, cross-attention and decoding pipeline while replacing the encoder by ViT-B and introducing Equation (4.1). Bone rotations and phenotypes are decoded by the same full-rig Anny configuration used to construct targets.

A DINOv2 ViT-L [ODM+23] with weights from NAVER's Anny checkpoint turns the image into tokens; one MLP scores each token for an anchor joint (the head, `--person_center head`) and a second predicts its 2D offset. Each detected token becomes a query of the HPH, which lets the people attend to each other and to all image tokens. The Anny head is what differs from the SMPL-X version: the rotations use the 6D representation [Zho+19] with Gram–Schmidt, the phenotypes a sigmoid, and the depth Multi-HMR's focal-normalised log-depth. Right after Gram–Schmidt four bones are multiplied by one constant rotation each (Section 4.6); then a fixed mask sets 88 helper bones (toes, breasts, face, subdivision bones) to the identity, in the prediction and the ground truth alike, and keeps 75. The Anny layer runs inside the network and places the body with Equation (4.2), so the 3D and 2D losses reach the network through it. The network has 318.38 M parameters (heads 14.01 M) and needs 1,927.4 GFLOPs per image with one person.

## 4.3. Supervised Training Objective

A loss penalizes a mismatch during training. Each term compares a prediction with its corresponding label: detection with detection, pose with pose, and so

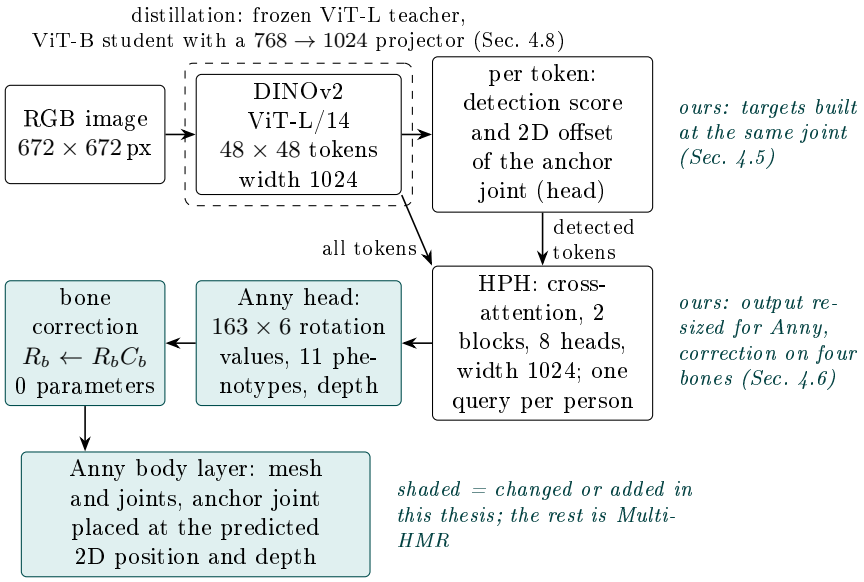



**Figure 4.2.:** *Teacher decoding architecture. A 672-pixel image passes through a DINOv2 ViT-L backbone (48 × 48 tokens). Every token gets a detection score and a 2D offset for the anchor joint, the head. For each detected person a cross-attention head (HPH) predicts 163 × 6 rotation values, 11 phenotypes and a depth; a constant correction turns four flipped bones back; the Anny layer builds the body and places it at the predicted distance. Shaded boxes are changed or added in this thesis; the dashed box marks the part that distillation replaces.*

on. Anny converts the labelled body parameters into target joints and vertices. All anchor targets refer to the same head joint (Section 4.5). The total loss is

$$\begin{aligned}\mathcal{L} = {} & \alpha_{\text{det}}\mathcal{L}_{\text{det}} + \alpha_{\text{off}}\mathcal{L}_{\text{off}} + \alpha_{\text{rot}}\mathcal{L}_{\text{rot}} + \alpha_{\text{shape}}\mathcal{L}_{\text{shape}} \\ & + \alpha_{\text{dist}}\mathcal{L}_{\text{dist}} + \alpha_{\text{t}}\mathcal{L}_{\text{t}} + \alpha_{\text{j3d}}\mathcal{L}_{\text{j3d}} + \alpha_{\text{v3d}}\mathcal{L}_{\text{v3d}} \\ & + \mathbb{1}[e \geq e_{\text{2D}}]\,(\alpha_{\text{j2d}}\mathcal{L}_{\text{j2d}} + \alpha_{\text{v2d}}\mathcal{L}_{\text{v2d}}) + \lambda_{\text{kd}}\mathcal{L}_{\text{kd}} + \lambda_{\text{kd-out}}\mathcal{L}_{\text{kd-out}},\end{aligned} \tag{4.3}$$

with $e$ the epoch; the distillation terms exist only for a student. Table 4.1 says what each term compares and gives its weights.

### 4.3.1. Detection and Anchor Localization

Let $s_q$ be the detection probability at token $q$, $y_q \in \{0, 1\}$ its binary target and $N_+ = \sum_q y_q$. The implemented focal loss is

$$\mathcal{L}_{\text{det}} = -\frac{1}{\max(1, N_+)} \sum_q \begin{cases} (1 - s_q)^2 \log(s_q + \epsilon), & y_q = 1, \\ s_q^2 \log(1 - s_q + \epsilon), & y_q = 0. \end{cases} \tag{4.4}$$

The sum covers the batch's token scores. Positive tokens represent head anchors; focal weighting reduces the contribution of easy background tokens. For $N_h$ supervised people and their sub-token offsets $\mathbf{o}_i$,

$$\mathcal{L}_{\text{off}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{o}}_i - \mathbf{o}_i\|_1. \tag{4.5}$$

| Term (log name) | Supervised quantity | Default | Final | Control | Student |
|---|---|---|---|---|---|
| Detection (`bce`) | focal loss on the per-token score "an anchor lies here" | 10 | 10 | 10 | 10 |
| Offset (`offset`) | L1 on the anchor's position inside its token | 1 | 1 | 1 | 1 |
| Rotations (`rotmat`) | L1 over the 9 entries of each predicted/target bone rotation | 0.1 | 0.1 | 0.1 | 0.1 |
| neck and hand boost | renormalised: $\times 2.41$ neck, head, hands; $\times 0.60$ rest | 1 | 4 | 4 | 4 |
| Shape (`shape`) | L1 on the 11 phenotypes after the sigmoid | 1 | 20 | 20 | 20 |
| Distance (`dist`) | L1 on the anchor's focal-normalised log-depth | 1 | 5 | 1 | 5 |
| Translation (`transl`) | L1 on the anchor's 3D position in the camera frame | 1 | 5 | 1 | 5 |
| 3D joints (`j3d`) | L1 on the 163 joints, each at its own pelvis | 100 | 100 | 100 | 100 |
| 3D vertices (`v3d`) | the same for the mesh vertices | 100 | 100 | 100 | 100 |
| 2D joints (`j2d`) | L1 in pixels on the joints projected from anchor and depth | 1 | 0.1 | 0.1 | 0.1 |
| 2D vertices (`v2d`) | the same for the projected vertices | 1 | 0.1 | 0.1 | 0.1 |
| first 2D epoch | `--start_2d_epoch` | 10 | 0 | 0 | 0 |
| Global rot. (`global_rot`) | L1 on camera-frame head, neck03, wrist orientation | 0 | 0 | 0 | 0 |
| Feature KD (`kd`) | KL of softened student and teacher tokens, $T = 4$ | – | – | – | 1.0 |
| Output KD (`kd_out`) | L1 to the teacher's rotations, phenotypes, depth | – | – | – | 1.0 |
| Target joint | `--gt_center_joint` (four anchor targets) | pelvis | head | pelvis | head |
| Bone correction | neck02, head, metacarpal1.L, finger4-1.R | none | on | none | on |

**Table 4.1.:** *Loss terms and their weights. "Default" is the value in `loss.py`; Final, Control and Student are the values of those runs (model names in Table 4.2). The total loss is the weighted sum of all rows above the rule. For people from 3DPW the rotation and global-rotation terms skip the 38 finger, thumb and metacarpal bones in all three runs; the mesh terms are not masked. The shape mask for 3DPW people is off in all three runs and on only in Final-masked (Section 4.7). The Student keeps the teacher's bone correction during training and is evaluated with its own, re-estimated after training (Section 4.8.1).*

The detection score is compared with the binary label, while the offset is compared with the target 2D offset. They are separate losses; the score and location are not compared with each other.

### 4.3.2. Pose, Shape and Placement

Let $\widehat{\mathbf{R}}_{ib}$ and $\mathbf{R}_{ib}$ denote predicted and target rotation matrices for person $i$, bone $b$. The supervised pose term uses entry-wise L1:

$$\mathcal{L}_{\text{rot}} = \frac{1}{N_h} \sum_i \sum_b m_{ib} w_b \|\widehat{\mathbf{R}}_{ib} - \mathbf{R}_{ib}\|_1, \qquad \mathcal{L}_{\text{shape}} = \frac{1}{N_h} \sum_i \sum_k |\widehat{\phi}_{ik} - \phi_{ik}|. \tag{4.6}$$

Here $m_{ib}$ suppresses unavailable 3DPW hand labels and $w_b$ is the normalized bone weight. Helper rotations are set to identity in predictions and targets. Phenotypes are sigmoid outputs; the final configuration supervises all phenotype entries, including the converter constants. Shape-mask variants are ablations of that choice. The implementation sums over bones or phenotypes before averaging people, so the weights in Table 4.1 depend on these reductions.

For predicted and target focal-normalized log-depth $d_i$ and camera-space anchor translation $\mathbf{t}_i$,

$$\mathcal{L}_{\text{dist}} = \frac{1}{N_h} \sum_i |\widehat{d}_i - d_i|, \qquad \mathcal{L}_{\text{t}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{t}}_i - \mathbf{t}_i\|_1. \tag{4.7}$$

These terms supervise absolute placement. They complement the centred geometric terms below.

### 4.3.3. Geometric and Reprojection Supervision

For joints or vertices $\mathbf{X}_i$, define pelvis-centred coordinates $\overline{\mathbf{X}}_{i\ell} = \mathbf{X}_{i\ell} - \mathbf{J}_{i,\text{root}}$. With $K_X$ points and optional normalized joint weights $w_\ell$,

$$\mathcal{L}_{X3D} = \frac{1}{N_h K_X} \sum_i \sum_{\ell=1}^{K_X} w_\ell \|\widehat{\overline{\mathbf{X}}}_{i\ell} - \overline{\mathbf{X}}_{i\ell}\|_1. \tag{4.8}$$

For vertices $w_\ell = 1$; joints use the configured weights. Subtracting each body's pelvis moves both pelvises to the origin. The comparison then checks the body around that common point, regardless of where it was placed in the scene. Placement therefore needs the separate losses in Equation (4.7). The 2D losses project the bodies back into the image without this pelvis alignment:

$$\mathcal{L}_{X2D} = \frac{1}{|\Omega_X|} \sum_{(i,\ell)\in\Omega_X} \|\pi_{\mathbf{K}}(\widehat{\mathbf{X}}_{i\ell}) - \pi_{\mathbf{K}}(\mathbf{X}_{i\ell})\|_1, \tag{4.9}$$

where $\Omega_X$ contains target points inside the image. The implementation activates the 2D losses at the configured epoch and suppresses non-finite reductions. The task objective is the weighted combination in Equation (4.3); the distillation additions are defined in Section 4.8.

The detection term is the focal loss of CornerNet [LD18] (the log calls it `bce`, but it is not a binary cross-entropy). The neck and hand boost moves rotation loss between bones, it does not add any. The shape L1 is smallest at the median of the plausible answers, so weak image cues pull predictions

toward the training median (Appendix F). The 3D terms align both skeletons at the pelvis, so only distance and translation see placement directly, and the 2D terms indirectly; before the depth fix the two direct terms made up about 2.5% of the loss, against 25.7% for shape. Training runs in fp32 with TF32 products (mixed precision gave NaN values with the Anny layer), and steps with NaN or Inf gradients are skipped (16 of 10,000 in Final). The main loss weights are listed in Table 4.1; the reported pilots test specific changes rather than an exhaustive weight search.

## 4.4. Training schedule and model names

**Teacher initialization.** The teacher uses a ViT-L backbone initialized from NAVER's released checkpoint and Anny-compatible prediction heads trained on AnnyOne. The synthetic-only Base checkpoint is the starting point for mixed-data teacher training. The earlier transfer stages are reproduction history, recorded in Appendix E.

**Final and Control.** Final starts from Base and trains 10 epochs of 1,000 iterations at batch 4 at a constant learning rate of $10^{-5}$; the last 4 backbone blocks and the heads train. Three changes come on together: the depth fix with distance and translation weight 5 (Section 4.5), the bone correction (Section 4.6), and 30% real images per batch with the hand mask (Section 4.7). The 55 images of the diagnostic manifest (Chapter 3) are never sampled. Training took 4,292 s on one A100. Control is identical but keeps the pelvis targets with weight 1 and has no bone correction, so Control against Final isolates the two fixes, and Base against Control shows the real images plus the restart cost. Each fix was tested alone only in short pilots (Appendix F).

**Model names.** Table 4.2 is the key to the names used from here on; each name is one run. Table A.1 in Appendix A gives each name's run, checkpoint, starting weights, data, schedule and flags.

| Name | Meaning |
|---|---|
| Multi-HMR-Anny | NAVER's released ViT-L model for Anny; external reference and source of the teacher backbone |
| Base | end of the schedule, AnnyOne only; the start of every fine-tune |
| Control | Base, 10 epochs, 30% 3DPW (corrected labels), hand mask, neither fix |
| Final | larger ViT-L teacher: Control plus consistent head anchors and bone correction |
| Final-masked | Final plus the shape mask on all 8 converter constants |
| Final-partial-mask | Final plus the shape mask on the 7 constants other than age |
| Student | deployed ViT-B model distilled from Final |

**Table 4.2.:** *Model names used in this thesis. Every name is one run; no run was repeated with another seed. Full definitions: Table A.1 (Appendix A).*

## 4.5. Consistent Head-Anchor Supervision

Detection, sub-token offset, depth and translation describe the same head joint used for mesh placement in Equation (4.2). The target builder first decodes the annotated body, selects that joint, and derives its camera-space location and image projection. Visibility checks and the one-person-per-token assignment use this anchor as well.

This consistency matters because pelvis-aligned geometry losses do not supervise absolute placement. If the target translation described a pelvis while the decoder placed a head there, the terms would disagree about the body's location. Figure 4.3 illustrates the distinction; Appendix F contains the supporting placement measurements. The final teacher and student use head-anchor targets. The distance and translation weights are 5, compared with 1 in the Control configuration. The ablation therefore changes anchor construction and placement weighting together.

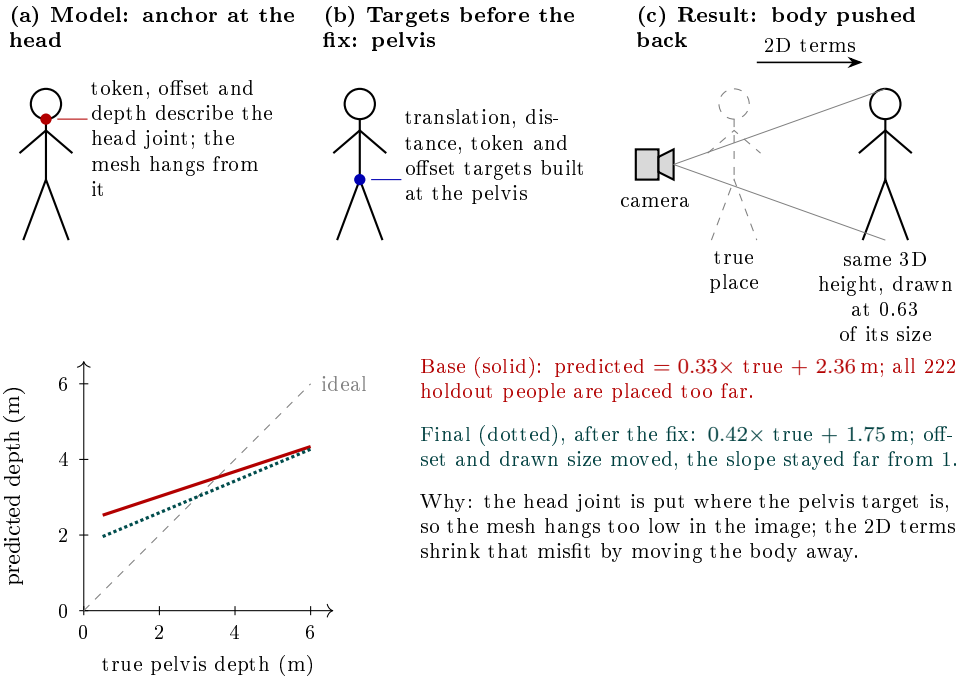



**Figure 4.3.:** *The predicted body is placed by its head joint, so detection, offset and depth targets must refer to that same point. The schematic contrasts inconsistent pelvis targets with head-anchor placement. The measured depth scatter is reported separately in Figure 5.6.*

## 4.6. Fixed Output Rotation Correction

The pose head's output includes a fixed calibration for selected bones before Anny decoding:

$$\mathbf{R}_{ib}^{\text{corr}} = \widehat{\mathbf{R}}_{ib}\mathbf{C}_b, \qquad \mathbf{C}_b = \text{Proj}_{SO(3)}\left(\frac{1}{N_{\text{cal}}}\sum_i \widehat{\mathbf{R}}_{ib}^{\top}\mathbf{R}_{ib}^{\text{target}}\right). \tag{4.10}$$

The projection finds the nearest proper rotation to the mean residual. Calibration uses people from 200 random training images, never evaluation images. Candidate bones have a persistent rotation offset above the configured threshold 90°. The teacher correction acts on neck02, head, metacarpal1.L and finger4-1.R. Figure 4.4 illustrates its geometric role.

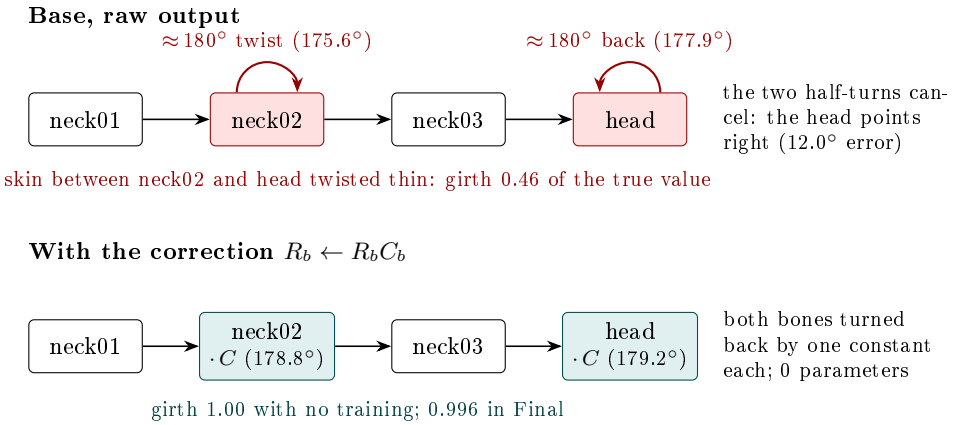



**Figure 4.4.:** *The neck chain neck01–neck02–neck03–head. Top: in Base, neck02 and the head bone are each turned about half a turn; the turns cancel, so the head points right, but the skin between them is twisted to 0.46 of its true girth. Bottom: one constant rotation per bone turns both back. Girth on the AnnyOne holdout, setting V1.*

The correction is applied before helper masking, Anny decoding and loss calculation. It introduces constants, not additional trainable weights. The teacher is fine-tuned with the correction in place; an unadapted post-processing change is not assumed to preserve global articulation. During student training, the copied heads use the teacher's correction. A correction is then estimated from the student's own training predictions and used for its evaluation and video inference.

A constant rotation can compensate a consistent residual but cannot resolve arbitrary image-dependent pose error. The supervised and distillation objectives remain responsible for articulated prediction. Appendix F reports per-bone, surface and global-orientation evidence and the limitations of interpreting each metric.

## 4.7. Training with real images

**The mix.** A weighted sampler draws on average 30% of each batch from 3DPW-train converted to Anny (Chapter 3; corrected labels, 5,882 images) and 70% from AnnyOne; with more labelled people per AnnyOne image, roughly 16% of the supervised people are real (an estimate). The test split is never used for training.

**The hand mask.** The converter fits only the root and the bones that carry a mapped SMPL joint, so every finger, thumb and metacarpal bone of a 3DPW label is at rest. With `--mask_3dpw_hands 1` the rotation terms skip these 38 bones for 3DPW people. Measured finger-rotation supervision therefore comes from AnnyOne. The mesh losses still include the fitted rest-pose hands, and the student's output distillation still uses the teacher's hand predictions. The mask does not remove all hand-related training.

**The shape mask, tested.** The converter leaves 8 phenotypes at constants (age 0.55; muscle, weight, cup size, firmness and three ethnicity values at 0.5). With `--mask_3dpw_shape 1` the shape term ignores them for 3DPW people (Final-masked); the partial variant masks only the 7 values other than age (Final-partial-mask). The mesh terms still carry the neutral body. I kept the unmasked model as Final and teacher, because both masked models lost the gender gain on real photos (Section 5.6.2).

## 4.8. Knowledge distillation

A frozen teacher supervises a smaller student on the same image, on top of the task loss (Figure 4.1).

**Feature distillation.** For image $u$ and token $n$, define channel distributions $p^t_{unc} = \mathrm{softmax}_c(\mathbf{z}^t_{un}/T)$ and $p^s_{unc} = \mathrm{softmax}_c(\widetilde{\mathbf{z}}^s_{un}/T)$. With batch size $B$, the implemented feature loss is

$$\mathcal{L}_{\mathrm{kd}} = \frac{T^2}{B} \sum_{u=1}^{B} \sum_{n=1}^{N_p} \sum_{c=1}^{C_t} p^t_{unc} \log \frac{p^t_{unc}}{p^s_{unc}}. \tag{4.11}$$

For each patch, softmax turns its feature channels into values that sum to one. Dividing by $T = 4$ makes this distribution less concentrated on the largest channels. KL divergence then penalizes differences between the teacher's distribution and the projected student's distribution. These are feature-channel distributions, not probabilities of body classes. The $T^2$ factor rescales the gradients [HVD15]. The implementation averages over images and sums over tokens and channels. Only the student and projector receive gradients from this term; the teacher stays fixed. Matching intermediate representations follows the idea of feature distillation [Rom+15].

**Head transfer and output distillation.** The student copies every teacher head tensor whose name and shape match. Both run on the true person locations, so person $i$ of one is person $i$ of the other, and

$$\mathcal{L}_{\text{kd-out}} = \frac{1}{N_h} \sum_i \Big[ \alpha_{\text{rot}} \sum_b \|\mathbf{R}^s_{ib} - \mathbf{R}^t_{ib}\|_1 + \alpha_{\text{shape}} \sum_k |\phi^s_{ik} - \phi^t_{ik}| + \alpha_{\text{dist}} |d^s_i - d^t_i| \Big]. \tag{4.12}$$

pulls the student toward detached teacher outputs, with the term weights of Table 4.1. The rotation distillation term sums all bone residuals without the supervised per-bone boost or missing-label mask. Teacher predictions are available even where a converted ground-truth hand rotation is not. For both distillation terms, the selected recipe uses $\lambda_{\text{kd}} = 1.0$ and $\lambda_{\text{kd-out}} = 1.0$.

**Why the copied heads are frozen first.** The heads already know how to turn teacher features into body predictions, but the new projector initially produces different inputs. I freeze the copied heads for the first 5 epochs so the encoder and projector can adapt without immediately changing those weights. I then train the heads too. This gives the student a useful starting point while allowing later adaptation to its own features.

### 4.8.1. The ViT-B student of Final

**Recipe.** The Student follows the projector-fed design with a ViT-B backbone and the corrected training of this chapter (Table 4.3). Its backbone starts from NAVER's Multi-HMR ViT-B, trained for mesh recovery, and trains fully; the copied heads stay frozen for 5 epochs; the targets are exactly the teacher's. The script contains a colour-jitter option, but the training code does not use it, so I do not count it as an applied augmentation. The copied heads output the half-turn that the teacher's bone correction undoes, so the student trains with that correction; afterwards its own is estimated on 200 training images and used in every evaluation.

**Accuracy-retention criterion.** In addition to reporting errors directly, the experiment records a retention criterion: for an error metric $E$, $E_s \leq E_t/0.95$. Table 4.3 also lists the thresholds against the released baseline. This is an explicit comparison criterion, not an assertion that the student achieves it. The teacher comparison uses the same body representation and evaluation convention. The baseline's AnnyOne result is potentially in-sample, and its 3DPW training exposure differs, so the two references must be interpreted separately.

## 4.9. Evaluation protocol

Table 4.4 lists the metrics. Errors are means over matched people: a greedy 2D matcher pairs each labelled person with at most one detection, and recall

| *A. Recipe* | | |
|---|---|---|
| | Final (the teacher) | Student |
| Backbone | ViT-L/14 (width 1024) from Base; last 4 of 24 blocks trained; frozen in bf16 when teaching | ViT-B/14 (width 768) from NAVER's HMR-trained Multi-HMR ViT-B; fully trained |
| Heads | HPH at width 1024 (14.01 M) | copied from the teacher, frozen 5 epochs, fed by a projector 768→1024 (0.79 M) |
| Params / GFLOPs / ms | 318.38 M / 1,927.4 / 59.7 | 101.38 M / 603.3 / 28.5 (3.14× / 3.19× / 2.10× less) |
| Schedule | 10 epochs × 1,000 iterations × batch 4 = 40,000 images | 30 epochs × 1,000 iterations × batch 8 = 240,000 images |
| Learning rate | $10^{-5}$, constant | $3 \times 10^{-5}$, halved every 12 epochs |
| Targets | head anchor, depth weight 5, 30% 3DPW (v23), hand mask | identical |
| Distillation | – | feature KL $\lambda = 1.0$, $T = 4$; output L1 $\lambda = 1.0$ |
| Bone correction | four bones, estimated on Base | teacher's in training; own one afterwards |
| *B. The 95% bars: student error ≤ reference error / 0.95* | | |
| | against Final | against Multi-HMR-Anny |
| AnnyOne holdout PVE (V1) | 73.7 → **77.6** | 60.3 → 63.5 (likely in-sample) |
| AnnyOne holdout MPJPE (V1) | 77.9 → **82.0** | 63.3 → 66.6 |
| 3DPW test PVE (v23) | 89.0 → **93.7** | 126.9 → 133.6 |
| 3DPW test MPJPE (v23) | 88.3 → 92.9 | 129.4 → 136.2 |

**Table 4.3.:** *The Student (ViT-B) next to its teacher Final. Part A: recipe and measured cost. Part B: the error bars of the 95% rule in mm; the job computes the bars from the teacher re-measured in the same job, so these values are for orientation. Cost at 672 px, one person; ms on one L40S, batch 1, both models in the same job. The rule also asks for fewer parameters, fewer FLOPs and a faster forward pass. Read as "error at most 1.05 times the teacher's", the bars against Final would be 77.4 and 93.4 mm. The Student's own errors are in Chapter 5.*

is the share of labelled people that got one.

For one matched person, pelvis-centred vertex and joint errors are

$$\mathrm{PVE} = \frac{1}{N_v}\sum_{v=1}^{N_v} \|\widehat{\mathbf{V}}_v - \overline{\mathbf{V}}_v\|_2, \qquad \mathrm{MPJPE} = \frac{1}{N_j}\sum_{j=1}^{N_j} \|\widehat{\overline{\mathbf{J}}}_j - \overline{\mathbf{J}}_j\|_2. \tag{4.13}$$

The bars denote subtraction of each body's pelvis. Scores are averaged over matched people and reported in millimetres. PA-PVE first fits one rotation, translation and uniform scale to align the predicted vertices with the target, then computes the mean Euclidean vertex distance. Recall is the number of

| Metric | What it measures | Blind to |
|---|---|---|
| PVE (mm) | vertex error after both pelvises are put at the same point | absolute placement and depth |
| PA-PVE (mm) | PVE after the best rotation, scale and shift (Procrustes) | also orientation and scale |
| MPJPE (mm) | PVE on the 163 Anny joints | as PVE |
| Recall, precision (%) | labelled people matched; detections matched | false detections; bystanders |
| Root error (m) | 3D pelvis distance, no alignment | pose and shape |
| Size ratio | drawn head-to-pelvis distance, predicted over true (1 is right) | 3D body size |
| Depth slope / intercept | line fitted to predicted against true pelvis depth (1 / 0 m) | errors around the line |
| Bone angle error (deg) | angle of $\hat{R}_b^\top R_b$ for a parent-relative rotation; ≈180 = flipped | compensating bones |
| Head direction (deg) | camera-frame head orientation error | the chain below the head |
| Neck girth ratio | neck-skin radius, predicted over true (1 is right) | neck length |
| $\mathrm{PVE}_{\mathrm{neck}}$ (mm) | vertex error over the neck skin, centred at neck01 | the rest of the body |
| Child AUC | chance that a child gets a lower predicted age than an adult (0.5 = chance) | undetected people |
| Gender accuracy (%) | predicted label using a threshold of 0.5 | undetected people |

**Table 4.4.:** *Metrics used in this thesis. All errors are means over matched people; the last two are on Relative Human; MPJPE is not comparable with published 3DPW numbers. On the AnnyOne holdout the headline setting is V1 (helper bones masked); V0 leaves them unmasked and reads 3.8 mm higher in PVE on the same people. 3DPW numbers are against our converted labels only (Chapter 3).*

matched labelled people divided by the total number of labelled people. These evaluation scores differ from the coordinate-wise L1 losses used during training. A lower centred error does not necessarily mean a better overlay in the camera image.

**AnnyOne holdout.** The last 100 AnnyOne images (indices 579,860 to 579,959) are excluded from training in this thesis and contain no children (Chapter 3). The released baseline may have seen them. The headline setting V1 masks the helper bones, as the demos do; V0 does not. A trace script adds the placement and per-bone metrics.

**3DPW test.** All 8,190 test images (11,854 people) of the corrected labels, same script for every model, setting V0. MPJPE averages all 163 Anny joints and the conversion itself errs by about 14 mm (Chapter 3), so these numbers are not comparable with published 3DPW numbers.

**Relative Human and demo images.** Relative Human gives real photos with an age group, a gender and a box per person. Its validation split (635 images, 2,130 people) serves for evaluation only: boxes from the projected vertices, matching at IoU $\geq 0.3$, child-shaped meaning a predicted age below 0.45. The 4 demo photos and 8 3DPW test frames are rendered with each body placed from its detection box, which hides depth errors, and are judged by eye.

**Training variation.** Each model has one recorded training run without a fixed training seed. The two identical short pilots differ by 7.6 mm PVE and 1.9° head direction (Section F.7). These observations motivate matched controls and caution with small differences, but do not provide confidence intervals or a significance threshold for every dataset and schedule. Repeated training of the final student and teacher was not performed. Evaluation of fixed checkpoints is kept separate from training variation.

## 4.10. Video Inference, Filtering and Alignment

The video application uses the trained image model without retraining it. It keeps the raw predictions and can add two steps: temporal filtering to reduce changes between frames, and landmark alignment to improve where the mesh is drawn. These steps address different problems.

**Raw prediction and tracking.** Frames are read in their original order and padded to each checkpoint's input size. Both models use the same camera assumption: by default, a 55° field of view along the longer image side. The raw mesh keeps its predicted camera-space placement. A geometric tracker follows one primary person using box overlap and centre distance. A missing detection remains missing; it is not replaced by an old mesh.

**Temporal filtering.** For shape values and the 3D head anchor, I use an exponential moving average:

$$\alpha_t = 1 - e^{-\Delta t/\tau}, \qquad \overline{\mathbf{x}}_t = (1-\alpha_t)\overline{\mathbf{x}}_{t-1} + \alpha_t \mathbf{x}_t. \tag{4.14}$$

Here $\mathbf{x}_t$ is the current prediction, $\overline{\mathbf{x}}_{t-1}$ is the previous filtered value and $\Delta t$ is the time between selected frames. A larger time constant $\tau$ gives stronger smoothing but can delay fast movement. Setting it to zero passes the current value through. The defaults are 1.0 s for shape, 0.12 s for the head anchor and 0.08 s for pose.

Bone rotations need a different interpolation because averaging their coordinates can produce an invalid rotation. I convert them to unit quaternions and use shortest-path spherical interpolation (SLERP) with the same $\alpha_t$. Anny then rebuilds the mesh from the filtered pose and shape and places it at the filtered anchor. The filter uses current and past outputs only. It resets on a missing detection, a track change or a timestamp gap above 0.25 s. No future frames or learned temporal weights are used.

**Landmark alignment.** MediaPipe [Goo] detects shoulders, elbows, wrists, hips, knees and ankles in the original video. These 2D landmarks are extracted separately and associated with the tracked person. The saved landmark file is checked against the source video's hash and dimensions to avoid using points from another video or an already rendered overlay.

I fit one translation $\boldsymbol{\Delta} \in \mathbb{R}^3$ to bring the projected Anny joints closer to these points. The optimizer uses confidence-weighted residuals and a robust loss, so one poor landmark has less influence. It requires confident points spanning the upper and lower body, rejects outliers and checks that the fitted body stays in front of the camera and within the allowed translation range. A rejected fit keeps the unaligned prediction. If filtering is enabled, alignment uses the filtered body.

This step changes position and depth, but leaves bone lengths, pose and shape unchanged. Moving a body farther away makes its projection smaller; moving it closer makes it larger. It can improve an overlay without proving that the body's 3D position is correct.

**Stabilizing aligned depth.** Independent alignment can choose a slightly different depth in each frame and make the mesh appear to grow and shrink. I therefore filter the logarithm of the fitted pelvis depth, using a default time constant of 0.35 s and limiting each log-depth step to $0.5\Delta t$. The whole body is translated along the current pelvis camera ray to reach this depth, which preserves the pelvis's image position. The state resets after rejected fits, missing predictions or track breaks. This is a trade-off: less size flicker may come with more current-frame landmark error. The reported projection error is recomputed from the mesh actually rendered.

**Measurements and timing.** For a pelvis-relative joint $\mathbf{q}_{tj} = \mathbf{J}_{tj} - \mathbf{J}_{t,\text{root}}$ and constant frame spacing $\Delta t$, the application records

$$\mathbf{v}_{tj} = \frac{\mathbf{q}_{tj} - \mathbf{q}_{t-1,j}}{\Delta t}, \qquad \mathbf{a}_{tj} = \frac{\mathbf{q}_{tj} - 2\mathbf{q}_{t-1,j} + \mathbf{q}_{t-2,j}}{\Delta t^2}. \tag{4.15}$$

It also records shape variation, missing frames and track segments. These are changes in predictions, not errors against ground-truth motion. Natural movement can increase them, and an incorrect body can still be very smooth.

Inference speed is computed as

$$\text{FPS}_{\text{infer}} = \frac{N_{\text{timed}}}{\sum_{t \in \mathcal{T}} \tau_t}, \tag{4.16}$$

where $\tau_t$ is synchronized elapsed time for the reported stage. The first ten frames are excluded from timing averages. Raw FPS covers the model forward pass; filtered FPS also includes filtering and Anny decoding; aligned FPS includes the active refinement steps. Preprocessing, rendering, video I/O and offline landmark extraction are excluded. These values differ from playback FPS and from complete online throughput. GPU, precision, resolution and person count must accompany a speed comparison.

# 5. Experiments and Results

I first compare the Student, its teacher (Final) and the released Multi-HMR-Anny baseline in accuracy and cost. I then describe the walking-video example. The remaining sections examine the teacher's anchor, bone and shape experiments, which explain the configuration used for distillation.

## 5.1. Evaluation sets and settings

The evaluation uses the datasets of Chapter 3 and the metrics of Section 4.9. The main synthetic result uses 100 held-out AnnyOne images with 222 people and no children. Setting V1 masks helper bones. The separate V0 setting leaves them unmasked, so its values should not be read as the same measurement. These images are excluded from thesis training, although the released baseline may have seen them.

The 3DPW test uses all 8,190 images of the corrected labels, in V0, with the same script for every model. These are errors against fitted Anny bodies, not published SMPL-space scores. Relative Human provides real-photo age groups and gender labels, but no reference 3D body. The 12 demo images are used for visual inspection; the historical demo renderer places bodies from detection boxes and therefore does not test predicted depth. The video renderer keeps camera-space placement.

Each thesis model has one training run. Final and Control start from Base and use the same schedule, real-image share and hand mask. Final also uses the anchor and bone corrections. The short repeated pilots provide a warning about training variation, not a significance threshold for these comparisons.

## 5.2. Compact Student and Reference Models

The Student has 101.38 M parameters, compared with 318.38 M for its teacher and 380.94 M for the released baseline (Table 5.2). Relative to the teacher it uses 3.19 times fewer FLOPs in the recorded configuration. The paired L40S benchmark gives 28.5 ms per student forward pass and 59.7 ms per teacher pass. The A100 and L40S columns are different measurements; latencies are compared only within the same hardware/protocol.

On converted 3DPW, Student PVE is 110.1 mm, between the teacher's 89.0 mm and the released baseline's 126.9 mm. On the synthetic holdout, the teacher also has lower error than the student. The released baseline's 60.3 mm synthetic result is potentially in-sample because the AnnyOne release includes this thesis's selected holdout images.

| | AnnyOne holdout (V1) | | | 3DPW test (v23) | | | |
|---|---|---|---|---|---|---|---|
| Model | PVE | PA-PVE | Recall | PVE | PA-PVE | Recall | 3DPW training |
| Released baseline | 60.3[a] | 44.6[a] | 89.0[a] | 126.9 | 68.8 | 97 | no |
| Teacher (Final) | 73.7 | 50.3 | 87.0 | **89.0** | **57.6** | 99 | yes |
| Student (ViT-B) | 119.6 | 80.8 | 85.4 | 110.1 | 71.8 | 99 | yes |
| Control | 77.4 | 56.4 | 89.0 | 91.5 | 59.7 | 99 | yes |
| Base | 70.5 | 53.3 | 90.2 | 121.8 | 72.0 | 100 | no |

**Table 5.1.:** *Reconstruction comparison. Errors are in mm; recall is in percent. One checkpoint is evaluated per model; thesis-trained models were not repeated across seeds. 3DPW errors use converted Anny labels and are not comparable with published SMPL-space results. [a]The released baseline was trained on the AnnyOne release, which includes these hold-out images, so its synthetic result is potentially in-sample. The real-image training exposure differs between released baseline and student/teacher.*

| Model | Params (M) | GFLOPs | Latency L40S (ms) | Latency A100 (ms) |
|---|---|---|---|---|
| Released baseline | 380.94 | ≥1917.2[b] | n/m | n/m |
| Teacher (Final) | 318.38 | 1927.4 | 59.7 | 77.9 |
| Student (ViT-B) | 101.38 | 603.3 | 28.5 | n/m |

**Table 5.2.:** *Model cost at 672 px input and one person. One multiply-add counts as two FLOPs. Latency is median batch-one forward time in fp32 with TF32; compare only within a GPU column. The student/teacher L40S pair was measured in the same job. n/m: not measured. [b]The baseline figure counts only its backbone, so total computation is higher. Base and Control share the teacher architecture; fixed rotation corrections add no trainable parameters.*

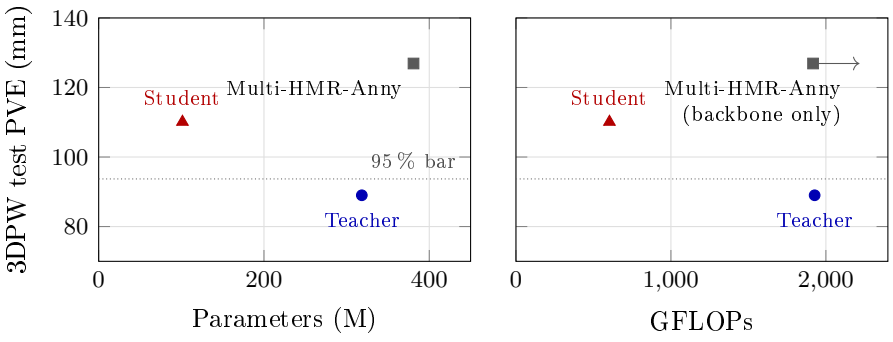



**Figure 5.1.:** *Model size and compute against 3DPW-test PVE (corrected labels, all 8,190 images; one run per model). Multi-HMR-Anny's GFLOPs count its backbone only (arrow: the true value is higher). Dotted line: the retention threshold against the teacher (93.7 mm).*

| Metric | Teacher | Student | Ratio | Limit | Met? |
|---|---|---|---|---|---|
| *AnnyOne holdout, V1* | | | | | |
| PVE (mm) | 73.7 | 119.6 | 1.62 | $\leq 77.6$ | no |
| PA-PVE (mm) | 50.3 | 80.8 | 1.61 | $\leq 52.9$ | no |
| MPJPE (mm) | 77.9 | 127.2 | 1.63 | $\leq 82.0$ | no |
| Recall (%) | 87.0 | 85.4 | 0.98 | $\geq 82.7$ | yes |
| *AnnyOne test-100, V0* | | | | | |
| PVE (mm) | 77.5 | 122.7 | 1.58 | $\leq 81.6$ | no |
| PA-PVE (mm) | 54.4 | 84.2 | 1.55 | $\leq 57.3$ | no |
| Recall (%) | 87 | 85 | 0.98 | $\geq 82.7$ | yes |
| *3DPW test (v23 labels), V0* | | | | | |
| PVE (mm) | 89.0 | 110.1 | 1.24 | $\leq 93.7$ | no |
| PA-PVE (mm) | 57.6 | 71.8 | 1.25 | $\leq 60.6$ | no |
| MPJPE (mm) | 88.3 | 112.0 | 1.27 | $\leq 92.9$ | no |
| Recall (%) | 99 | 99 | 1.00 | $\geq 94.1$ | yes |
| *Cost (L40S, batch 1, one person)* | | | | | |
| Params (M) | 318.38 | 101.38 | 3.14× | fewer | yes |
| GFLOPs | 1927.4 | 603.3 | 3.19× | fewer | yes |
| ms per image | 59.7 | 28.5 | 2.10× | faster | yes |

**Table 5.3.:** *The Student against its teacher (Final) and the 95 % rule: every error of the Student at most the teacher's divided by 0.95, every recall at least 0.95 of the teacher's, and fewer parameters, fewer FLOPs and a faster forward pass. The rule fails on every error and holds on recall and cost. One run, no seed. Against Multi-HMR-Anny the PVE bars would be 63.5 mm on AnnyOne (potentially in-sample for it; not met) and 133.6 mm on 3DPW (met). Cost: both models timed in the Student's job on one L40S.*

The external 3DPW comparison includes a training-data difference. Both student and teacher use converted 3DPW training images; the released baseline is evaluated without that adaptation. The result demonstrates the behaviour of the complete trained systems, rather than isolating network architecture. The measured cost/accuracy relation is visualized in Figure 5.1.

The historical cost benchmark uses median forward latency at batch one and a specified one-person configuration. The later per-video application measures mean synchronized latency over actual frames. Those protocols must remain separate; a video FPS result cannot be inferred from playback speed.

## 5.3. Reconstruction Accuracy and Distillation

The Student uses the training recipe in Table 4.3 and runs for 30 epochs. Its synthetic PVE is 119.6 mm, compared with 73.7 mm for the teacher; its 3DPW PVE is 110.1 mm, compared with 89.0 mm. These are 1.62 and 1.24 times the teacher's errors. MPJPE and PA-PVE also miss the retention thresholds, while recall meets them (Table 5.3).

Figure 5.2 helps explain what a longer run might change. Synthetic V0 error falls from 139.5 mm at epoch 15 to 122.8 mm at epoch 30. The 3DPW

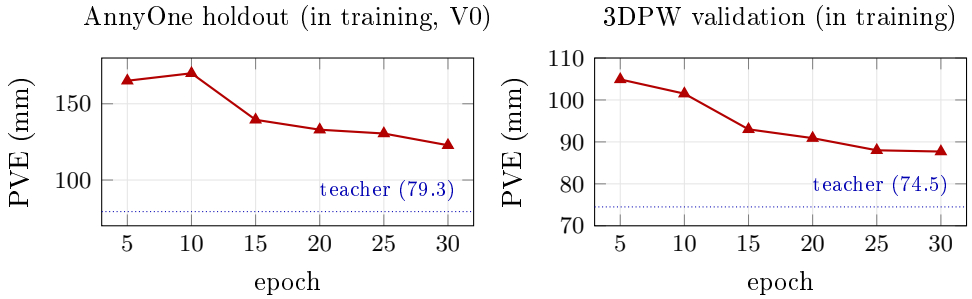



**Figure 5.2.:** *The Student's PVE during training, every five epochs: left the AnnyOne holdout as evaluated during training (setting V0), right the corrected 3DPW validation labels. Dotted: the teacher's value at the last epoch of its own training (Figure D.6). The heads were frozen in the first five epochs. One run, no seed; values from the training log.*

validation error changes only from 88.0 to 87.7 mm in the last five epochs. More training may improve the synthetic result, but the nearly flat real-image curve suggests that training duration is not the only limitation. A longer completed run is needed to test this.

Under the stated 95 % retention criterion, the student fails against the teacher on both sets. Its PVE is about 62 % higher on AnnyOne and 24 % higher on 3DPW, while the criterion allows about 5 %. Against the released baseline, it improves 3DPW PVE by 16.8 mm and meets the baseline's 3DPW threshold of 133.6 mm. It misses the synthetic threshold of 63.5 mm, which is based on a potentially in-sample baseline result.

The student is therefore useful as a lower-cost model, but this experiment does not show that it retains teacher-level accuracy. The baseline comparison also includes different training exposure. I did not repeat the student run or separately evaluate its fine hand and head articulation on labelled real images.

## 5.4. Video Application

I first ran the Student and released baseline on the same `man_walking.mp4` sequence without smoothing. Both checkpoints loaded completely, the NVIDIA L40S renderer passed its test, and both models processed all 356 frames. Table 5.4 reports this original run.

| Recorded quantity | Student | Released baseline |
|---|---|---|
| Processed frames | 356 | 356 |
| Frames with a matched person | 356 | 354 |
| Frames without a matched person | 0 | 2 |

**Table 5.4.:** *Coverage in the original unsmoothed walking run. A matched frame means that the tracker selected a person. It is not detection recall against labelled people or a 3D accuracy score. These counts do not measure the later filtering and alignment settings.*

Inspection of the later smoothed example showed why stability and alignment need separate checks: the movement looked smoother, but the mesh was displaced from the person. I added the landmark fit described in Section 4.10. The aligned example fitted the image more closely by eye, but its apparent size changed as the person walked. This led to the additional log-depth filter. These observations come from one clip and do not provide a measured improvement in PVE or temporal error. A completed comparison with the depth filter's numerical summaries is still needed.

The current tool can show five panels: the source frame, raw student, filtered student, aligned student and raw baseline. This separates the network prediction from the changes made afterwards. All student panels share the same forward pass. The baseline panel remains unfiltered, so a smoother student panel alone cannot establish that the student network has better temporal predictions.

The application records timings for the individual stages and saves pose, shape and tracking summaries. This report retains the verified paired student/teacher benchmark as its numerical speed comparison. It does not combine FPS displayed by different video revisions with that benchmark. Offline landmark extraction would also need to be included when assessing complete online throughput.

## 5.5. Teacher Supervision Ablations

The following comparisons analyse the larger teacher used for distillation. They do not independently measure the same anatomical improvements in the deployed student.

### 5.5.1. Reconstruction on AnnyOne

Table 5.5a compares Final with Control, Base and the two pilots. Final has the lowest PA-PVE among these teacher variants, 6.1 mm below Control. Its PVE is 3.7 mm below Control, while Base keeps the lowest PVE in this group. Final's recall is 2.0 percentage points below Control and 3.2 points below Base.

Figure 5.3 shows the placement difference: Base tends to draw smaller bodies farther from the camera, while Final is closer to the reference in these examples. PVE cannot capture that whole-body translation because it first aligns the pelvises. The training curves (Figure D.6) put Final below Control at all ten 3DPW validation epochs, while their synthetic V0 order changes between epochs.

The single runs do not establish whether the small PVE difference is repeatable. The recall decrease is an observed cost of the final configuration. Moving the detection target to the head may affect people whose heads are hidden, but I did not test that explanation. The matched sets also differ (214 for Final, 219 for Control and 222 for Base), so a comparison restricted to the same detected people would be useful.

| | Base | Depth pilot | Neck pilot | Control | Final |
|---|---|---|---|---|---|
| PVE (mm) | 70.5 | 74.8 | 68.8 | 77.4 | 73.7 |
| PA-PVE (mm) | 53.3 | 54.0 | 50.9 | 56.4 | **50.3** |
| Recall (%) | 90.2 | 87.8 | 89.4 | 89.0 | 87.0 |
| Root error (mm) | 1208 | 544 | 1429 | 1402 | **728** |
| Depth slope | 0.33 | 0.38 | 0.34 | 0.28 | 0.42 |
| Depth intercept (m) | 2.36 | 1.59 | 2.56 | 2.66 | 1.75 |
| Size ratio | 0.63 | 0.84 | 0.58 | 0.59 | 0.78 |
| Neck girth pred/GT | 0.461 | 0.457 | 0.997 | 0.464 | 0.996 |
| $PVE_{neck}$ (mm) | 27.3 | 27.7 | 10.7 | 27.7 | 11.0 |
| Global head direction (deg) | 12.0 | 12.7 | 14.4 | 13.0 | 13.1 |
| Fingers (deg) | 17.3 | 17.4 | 8.3 | 17.4 | 8.3 |
| neck02 bone (deg) | 175.6 | 175.5 | 4.7 | 175.8 | 4.8 |
| head bone (deg) | 177.9 | 178.0 | 11.2 | 177.7 | 8.9 |

**(a)** *AnnyOne holdout, 100 images, setting V1 (helper bones masked, matched people); root error, depth fit and size ratio with each body at the model's own predicted depth.*

| | Base | Mixed-L | Neck pilot | Control | Final | Multi-HMR-Anny |
|---|---|---|---|---|---|---|
| PVE (mm) | 121.8 | 95.5 | 120.2 | 91.5 | **89.0** | 126.9 |
| PA-PVE (mm) | 72.0 | 62.9 | 68.7 | 59.7 | **57.6** | 68.8 |
| MPJPE (mm)[b] | 126.9 | 94.8 | 124.8 | 90.9 | **88.3** | 129.4 |
| Precision (%) | 54 | 73 | 53 | 62 | 64 | 75 |
| Recall (%) | 100 | 94 | 100 | 99 | 99 | 97 |
| AnnyOne test-100 PVE[c] | 74.3 | 78.5 | 72.7 | 81.2 | 77.5 | (60.3)[a] |

**(b)** *3DPW test, corrected (v23) labels, all 8,190 images, setting V0, same evaluation script for every model. Mixed-L was trained on the labels before their correction and is tested on the corrected ones, like every model here.*

**Table 5.5.:** *Final against its Control and against Base. One run per model, no seed; two identical runs differed by 7.6mm PVE, so the PVE gap between Control and Final (77.4 vs 73.7) is within run-to-run variation, while PA-PVE, root error, neck and fingers are not. Depth pilot: depth fix, depth weight ×5, 10 epochs; neck pilot: bone correction and a 2-epoch fine-tune. 3DPW numbers are against our converted labels and are not comparable with published 3DPW results. [a]Multi-HMR-Anny masks the helper bones by design, so its number is V1 and compares with Final's 73.7 in (a); its exposure to these 100 images is unverified, so the result is potentially in-sample. [b]Mean over all 163 Anny joints, pelvis-centred. [c]The same 100 images as (a) in setting V0 (helper bones unmasked).*

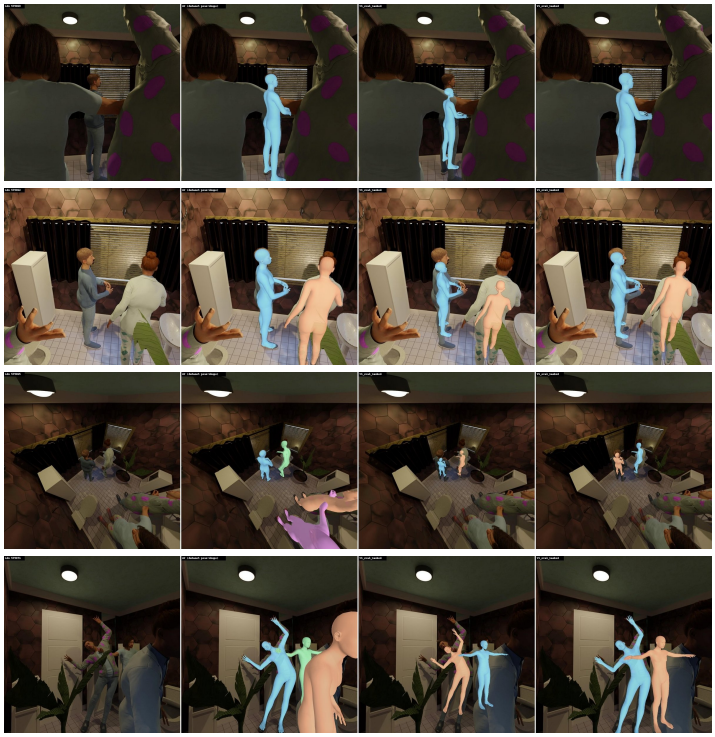


**Figure 5.3.:** *Four AnnyOne holdout images (index in the first panel), each as image | ground truth | Base | Final. Ground truth drawn with the dataset camera; Base and Final in setting V1 (panels labelled "V1_eval_masked"), each body at its model's own predicted depth. By eye, Base draws its bodies smaller and further back than the ground truth, Final close to its size and place. Mesh colours mark detections, not identities. People without a ground-truth mesh (the foreground arm in 579862) are not counted; people a model misses (the two lying in the foreground of 579865) have no mesh. Synthetic images, one run per model.*



(a) `downtown_bar` (b) *weekly market*

(c) `office_phoneCall` (d) *guitar*

**Figure 5.4.:** *Four 3DPW test frames, each shown as photo and Base (left half), then photo and Final (right half; the labels in the image name the two checkpoints). Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The other four demo frames are in Figure D.4 (Appendix D).*

### 5.5.2. Real-Image Supervision

Table 5.5b scores every model on all 8,190 test images with the same script. Final has the lowest error on our 3DPW-Anny labels, 89.0 mm PVE, ahead of Control, Mixed-L (the previous best), Base and Multi-HMR-Anny, and finds 99 % of the labelled people. Precision is low for every model. By eye (Figure 5.4), most men get male bodies in Final and female-looking ones in Base, Final's bodies are a little larger, and the seated guitar player is poor in both.

Most of the gain over Base comes from training on 3DPW at all: Control, without either fix, is 30.3 mm better than Base. The fixes add 2.5 mm PVE and 2.1 mm PA-PVE on top. Repeat variability on 3DPW was not measured, so these small single-run differences do not establish statistical significance. Final is 6.5 mm better than Mixed-L, which saw the old labels; whether the corrected labels cause part of this was not tested, since Mixed-L also differs in schedule. Precision is low because 3DPW labels only one or two actors per sequence, so every detected bystander counts as a false positive, as do ghost bodies on half-visible people.

I did not compute a sequence-level breakdown. The visual examples cover eight test frames and cannot represent every pose or scene. The converted-label scores also do not support a direct ranking against published SMPL-space 3DPW results.

### 5.5.3. Camera-Space Placement

Figure 5.5 shows the bug and the fix on one image: Base draws the man far behind his true position and too small, the depth pilot close to his true size.

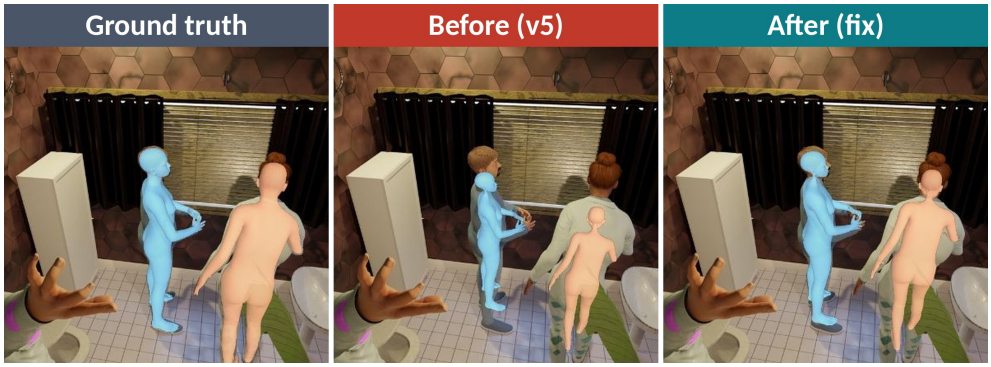



**Figure 5.5.:** *Holdout image 579862 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). The predicted body goes from 0.76 to 0.95 of its true height. Synthetic image; a good case, not a typical one. Image 579860 is in Figure D.1.*

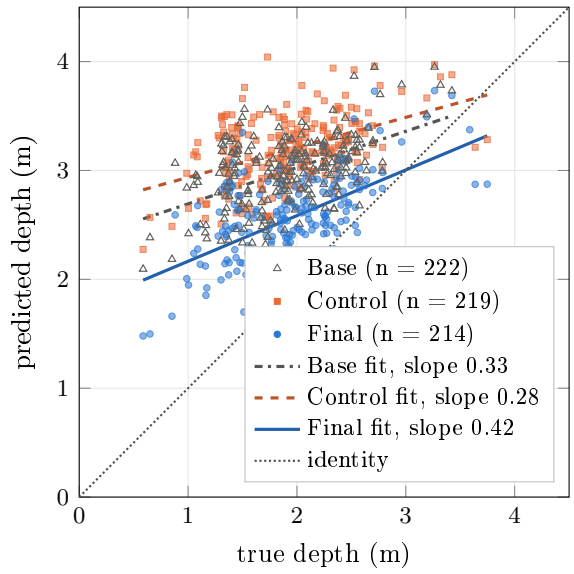



**Figure 5.6.:** *Predicted against true root depth for every matched person of the AnnyOne holdout (100 images, setting V1, each body at its own predicted depth): Base (222 people), Control (219) and Final (214). Lines: least-squares fits and the identity. Final's fit is* $0.42 \times \text{true} + 1.75$ *m, Control's* $0.28 \times \text{true} + 2.66$ *m, Base's* $0.33 \times \text{true} + 2.36$ *m. One run per model; synthetic images. Statistics in* `data/plot_stats.md`.

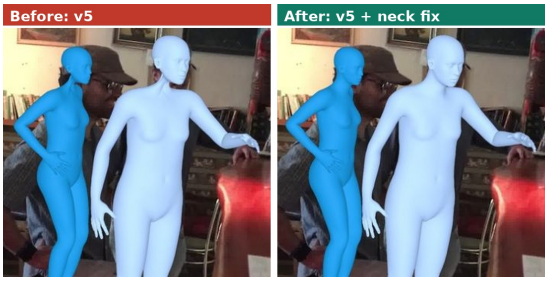



**Figure 5.7.:** *Real 3DPW test frame* `downtown_bar_00` *701: Base (left) and the neck pilot, fine-tuned 2 epochs with the correction (right); necks ringed. Bodies placed from the detector box; judged by eye. The columns differ by the correction and the fine-tune. A second frame is in Figure D.3.*

Figure 5.6 shows all people. Control places 99.1 % of them too far. Final reduces this bias: the root error falls by 48 % (1,402 to 728 mm), the drawn size rises from 0.59 to 0.78, the depth correlation from 0.46 to 0.56, and the head lands within 7.6 pixels of its true position against 67.1 (Table 5.5a).

The fix survives ten epochs with real images and the bone correction: every run with head targets, pilots included (Table F.3), lies below every run without them. What moved is the offset and the size, not the slope. Final's fit crosses the identity at 3.01 m, so near people are still placed too far (94.4 % of all people). The pelvis lands farther from its pixel (41.1 against 26.7 pixels), likely because the head is pinned while the body is still too small; I did not check this.

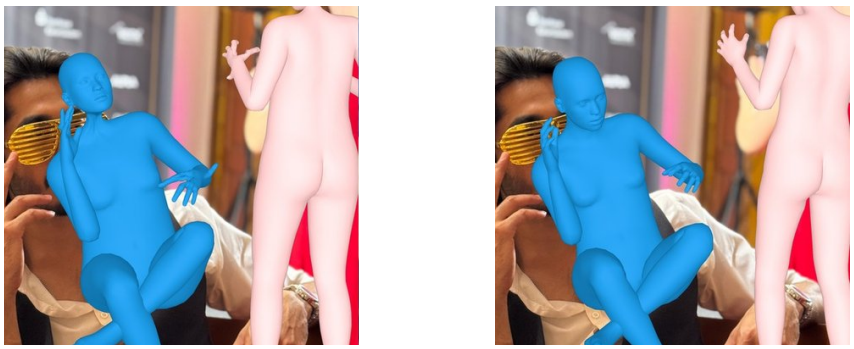
Near and far placement remain imperfect, and these tests do not explain the remaining slope error. Final's root error is above the 10-epoch depth pilot's 544 mm. Mixed data and bone correction also differ between those runs, so I cannot assign a cause. All depth measurements here come from synthetic images; Figure 5.5 illustrates the pilot.

### 5.5.4. Bone Correction

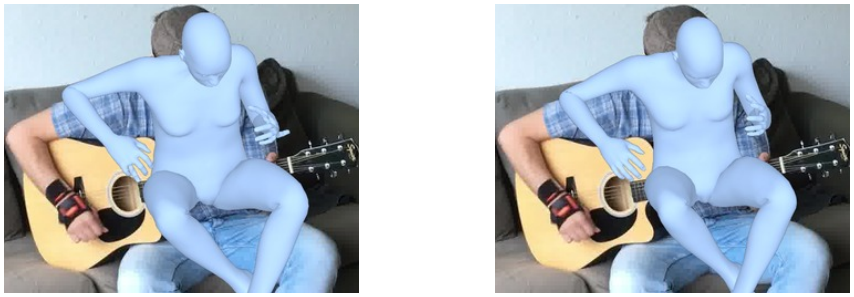
Figure 5.7 shows the correction on a real frame that no model trained on: Base draws both necks as thin stalks under the head, the corrected model draws them at normal width. Figure F.1 and Table F.4 give the correction without training and after a 2-epoch fine-tune. Final keeps it through ten epochs with real images (Table 5.5a): neck02 is 4.8 deg off against 175.8 for Control, the neck girth 0.996 of the truth against 0.464, and the joint error of the neck region falls from 62.6 to 5.5 deg. The head direction, which got worse with the correction alone (19.1 deg), is back at Control's level.

The half-turn does not come back during training: the network learns on top of the correction instead of undoing it. The correction is one constant rotation per bone, adds no trainable parameters and says nothing about why the bones flipped. The whole-body PVE comparison does not resolve a separate cost of this correction (Section 5.5.1).

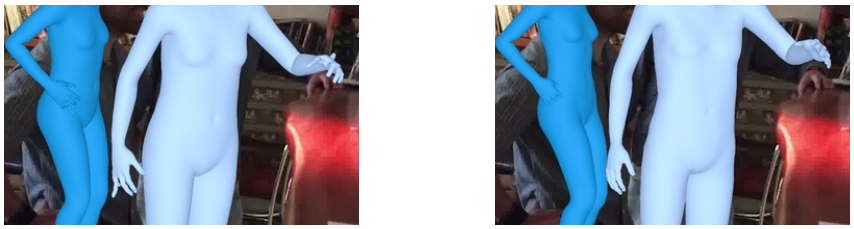
Base, predicted fingers | Final, predicted fingers



(a) party photo: the hand at the face and the hand over the armrest



(b) 3DPW test frame, guitar: strumming hand (left) and fretting hand (right)



(c) 3DPW test frame `downtown_bar`: the hand on the bar and the hanging hand

**Figure 5.8.:** *Predicted fingers (not set to rest pose) on three demo images, zoomed on the hands; Base left, Final right. Bodies placed from the detector box, so meshes sit beside their people. Base still carries the two flipped hand bones; Final has the correction. Neither model puts the fingers on the real ones. Three images, judged by eye; the columns differ by the correction and the fine-tune. All twelve montages are in Figure D.7.*

The real photos have no reference neck geometry, so I cannot report a real-image neck error. The pictured neck pilot also differs from Base in both correction and fine-tuning. The synthetic control retains a girth ratio of 0.461 without correction (Table F.4), which supports the correction's role. I did not test a more general two-sided correction for compensating rotations along the chain.

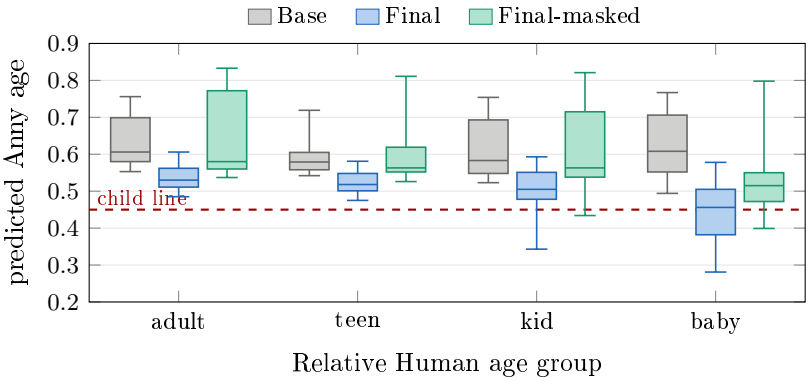



**Figure 5.9.:** *Predicted Anny age per Relative Human age group for Base, Final and Final-masked, matched (detected) people of the validation split only. Box: quartiles and median; whiskers: 5th and 95th percentile. Dashed: the child line, age 0.45. Ages are Anny phenotype values, not years. One run per model; statistics in* `data/plot_stats.md`.

### 5.5.5. Hand Articulation

Predicted fingers are 17.3 deg off in Base and 17.4 deg in Control, worse than fingers left at rest (15.1 deg; Table F.5). Final, with the two flipped hand bones corrected, reaches 8.3 deg (Table 5.5a); the correction alone, without training, gave 8.2.

About half of the finger error was two flipped bones, metacarpal1.L and finger4-1.R, and ten epochs of training keep the gain. The remaining error is not explained by these tests. The 3DPW labels provide no measured finger rotations, so their 38 hand bones are masked in the supervised rotation loss. Other mesh terms still include the fitted hands, and student distillation can use the teacher's hand outputs.

Figure 5.8 shows predicted fingers on real images; every earlier demo set them to rest pose. By eye, Base's fingers are splayed and stiff, with one finger of the left hand sticking out as the flipped metacarpal1.L would make it, while Final's are closed and bent and read as hands: draped over an armrest, curled as if fretting, bent over a bar edge. Where the hands are small (Figure D.7), nothing can be judged at print size.

These examples do not establish finger accuracy on real images. The 3DPW targets contain rest-pose hands, and the demo photos have no finger labels. The three visible examples show correction and fine-tuning together.

## 5.6. Shape Generalization of the Teacher

### 5.6.1. Children and Gender on Real Images

Table 5.6 measures body shape on real photos with labels. Final finds more people than Base and gets the gender right for 64.2 % of them against 46.5 %. Women are recognised (mean predicted gender 0.861, where 1 is female); men

| | Base | Final | Final-masked | Final-partial-mask |
|---|---|---|---|---|
| Recall (%) | 47.1 | 64.7 | 57.2 | 63.0 |
| Gender accuracy (%) | 46.5 | 64.2 | 49.3 | 47.4 |
| mean gender, men (male = 0) | 0.535 | 0.551 | 0.717 | 0.725 |
| mean gender, women (female = 1) | 0.551 | 0.861 | 0.860 | 0.888 |
| Child-vs-adult AUC | 0.606 | 0.719 | 0.708 | 0.727 |
| Mean age, adult | 0.631 | 0.536 | 0.638 | 0.532 |
| teen | 0.597 | 0.524 | 0.616 | 0.515 |
| kid | 0.613 | 0.500 | 0.611 | 0.506 |
| baby | 0.624 | 0.438 | 0.524 | 0.465 |
| Babies with age $< 0.45$ (%) | 2.6 | 49.3 | 16.9 | 34.3 |
| Mean height, adult | 0.507 | 0.500 | 0.505 | 0.527 |
| kid | 0.451 | 0.466 | 0.461 | 0.497 |
| baby | 0.394 | 0.464 | 0.478 | 0.504 |
| Matched children + adults | 948 | 1,305 | 1,157 | 1,270 |
| *Demo images: standard deviation of the prediction* | | | | |
| Muscle | – | 0.04 | 0.06 | n/m |
| Weight | – | 0.04 | 0.11 | n/m |
| Cup size | – | 0.05 | 0.11 | n/m |
| Firmness | – | 0.04 | 0.11 | n/m |
| Age | – | 0.03 | 0.08 | n/m |

**Table 5.6.:** *Body shape on real photos. Upper part: Relative Human validation split (635 images, 2,130 labelled people), detected people only, matched to the labelled boxes at IoU $\geq 0.3$; ages and heights are Anny phenotype values in $[0, 1]$, not years; "child-shaped" means a predicted age below 0.45; the AUC ranks kids and babies against adults by predicted age (0.5 = chance). The matched sets differ between models because recall differs. Lower part: spread of the predicted phenotypes over all detected people of the 12 demo images (4 photos and 8 3DPW test frames), not matched to people; not reported for Base, not measured (n/m) for Final-partial-mask. Final-masked masks all eight constant 3DPW phenotypes, Final-partial-mask the seven at 0.5 (age kept supervised; Section 5.6.2). One run per model, no seed. The neck pilot (bone correction and a 2-epoch fine-tune) is the same as Base within run-to-run variation (AUC 0.613).*

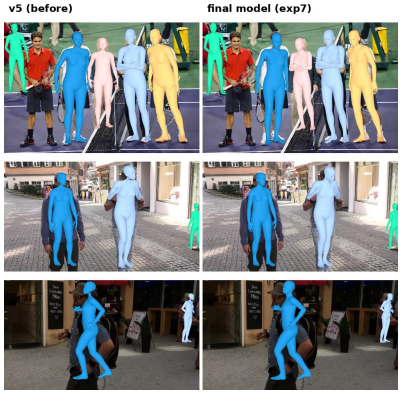



**Figure 5.10.:** *Three real photos never used in training (a tennis photo and the 3DPW test frames* `downtown_walking_00` *and* `downtown_runForBus_00`*): Base left, Final right, bodies placed from the detector box. Most men get male bodies, women stay female, the boy stays an adult body. Judged by eye; no metric exists for these images.*

stay in the middle (0.551). Children separate from adults: the AUC rises from 0.606 to 0.719, and 49.3 % of the babies are child-shaped against 2.6 % for Base. In Figure 5.9 Base's boxes stand at the same height for every group, Final's step down from adults to babies, and only the baby box reaches below the child line. On the demo photos (Figures 5.10 and D.5) most men look male in Final and the boy keeps an adult body.

The AUC ranks people by predicted age, so a common shift of all ages cannot raise it: the step is real separation of children from adults. Part of the lower ages is likely such a shift from the 3DPW age label, which the converter fixes at 0.55 for every real person. The gender gain fits the corrected 3DPW gender labels, which supervise the shape head on real people for the first time. The age-group scores improve, but child proportions remain unreliable. Male-labelled examples also retain intermediate values of Anny's gender parameter.

I report the Anny age parameter rather than age in years because the available age calibrations disagree. The synthetic holdout contains no children. No labelled real-image evaluation of the remaining body-build parameters is available, and each model's scores cover its own detected people.

The rest-pose line-up of Figure 5.11 draws every detected person in rest pose, so only the body differs. By eye, Base gives nearly everyone the same slim, flat body. Final gives the singers a fuller chest and hips and the man in the bar a broader torso. The tennis men keep a slim body with a slight chest in all three models, in line with the men's mid-range gender score. Final-masked looks like Final except for a slimmer man in the bar.

### 5.6.2. Shape-Supervision Masking

Two runs test the shape mask of Section 4.7, one run each. Final-masked stops the shape loss on the 8 values the converter fixes for every 3DPW person: age




(a) tennis photo: five men in a line



(b) stage photo: three singers



(c) 3DPW test frame `downtown_bar`: two men

**Figure 5.11.:** *Rest-pose line-up on three demo images: every detected person drawn in rest pose over the photo, so only body shape differs; Base, Final and Final-masked from left to right. Bodies placed from the detector box; judged by eye, one run per model. The other nine images are in Figure D.8.*

| | Final | Final-masked | Final-partial-mask | Rule |
|---|---|---|---|---|
| Masked 3DPW phenotypes | none | 8 (age + 7) | 7 (age kept) | |
| *AnnyOne holdout, V1* | | | | |
| PVE (mm) | 73.7 | 71.7 | 73.0 | |
| PA-PVE (mm) | 50.3 | 50.7 | 51.2 | |
| Recall (%) | 87.0 | 86.2 | 87.4 | |
| Root error (mm) | 728 | 682 | 680 | |
| *3DPW test, v23 labels, V0* | | | | |
| PVE (mm) | 89.0 | 91.1 | 89.9 | |
| PA-PVE (mm) | 57.6 | 59.0 | 58.9 | |
| *Relative Human validation* | | | | |
| Recall (%) | 64.7 | 57.2 | 63.0 | |
| Gender accuracy (%) | 64.2 | 49.3 | 47.4 | $\geq 60$ |
| Child-vs-adult AUC | 0.719 | 0.708 | 0.727 | $\geq 0.7$ |
| Babies with age $< 0.45$ (%) | 49.3 | 16.9 | 34.3 | |
| *Demo images, standard deviation of the prediction* | | | | |
| Muscle | 0.04 | 0.06 | n/m | $> 0.08$ |
| Weight | 0.04 | 0.11 | n/m | $> 0.08$ |

**Table 5.7.:** *The three shape-mask arms. Final supervises all converted 3DPW phenotypes; Final-masked masks the 8 constants of the converted labels (age 0.55 and seven values at 0.5); Final-partial-mask masks only the seven at 0.5 and keeps age supervised. Gender stays supervised in all three. AnnyOne holdout: 100 images, setting V1; 3DPW test: all 8,190 images, corrected labels, setting V0; Relative Human: validation split, matched people (full columns in Table 5.6); demo: 12 images, not matched to people; n/m: not measured. Rule: the thresholds, fixed before the first masked run, that a masked arm had to meet to replace Final. One run per arm, no seed; two identical runs differed by 7.6 mm PVE.*

0.55 and seven values at 0.5. Final-partial-mask stops it only on the seven and keeps age supervised. Gender stays supervised in both. Table 5.7 sets the two next to Final. On Relative Human, Final-masked loses most of Final's gain (gender 49.3 %, babies child-shaped 16.9 %), but on the demo images it brings back variety: the spread of weight, cup size and firmness rises from 0.04–0.05 to 0.11. Final-partial-mask keeps the children (AUC 0.727, 34.3 % of babies child-shaped) and loses gender as the full mask does, with both sexes on the female side (men 0.725, women 0.888).

The body errors of both arms remain close to Final; repeated runs would be needed to interpret small differences. The 3DPW mesh targets keep the neutral body; the trade-off is in the shape head. Without the mask, Final learned the seven constants and predicts nearly the same muscle, weight, cup size and firmness for everyone on real photos. The partial mask separates the two parts of the loss: keeping age supervision also preserves the age-group improvement in this run. This suggests that the constant age target affects the result, but does not establish the full mechanism. Gender falls to chance although its label stays supervised. A reading that was not tested: Final's gender gain depends on the seven constants on the 3DPW men as much as on the gender label.

Before the first masked run I fixed a rule: a masked model would replace Final if gender accuracy stayed at 60 % or more, the AUC at 0.7 or more, and the spread of muscle and weight rose above 0.08. Both masked runs fail the gender condition while passing the AUC condition, so I keep Final as the Student's teacher.

A wider spread of predicted values does not show that those values are correct: the available real-image labels do not measure muscle or weight. Final-partial-mask also has no comparable demo run, and the reason for its lower gender accuracy remains unresolved.

# 6. Discussion

The results show that the student reduces model cost, but does not keep the teacher's reconstruction accuracy. They also show why label quality, body geometry and mesh placement need separate checks.

## 6.1. Answers to the Research Questions

**Real-image supervision.** The converter makes 3DPW usable for Anny training. Its checked body joints differ from the source joints by 14.1 mm at the median. It fits pose, height and proportions, but it cannot recover information that the source joints do not provide. In particular, 8 shape values are fixed, and many hand and head rotations remain at rest.

Adding real images gives the largest improvement on converted 3DPW. Control reaches 91.5 mm PVE, compared with 121.8 mm for the synthetic-only Base. Adding the anchor and bone corrections gives Final's 89.0 mm. This comparison supports mixed training, but does not separate every effect of the data, restart and corrections.

**Compact reconstruction.** The student uses 3.14 times fewer parameters and 3.19 times fewer FLOPs than the teacher. Its paired L40S forward pass is 2.10 times faster. However, PVE rises from 73.7 to 119.6 mm on AnnyOne and from 89.0 to 110.1 mm on 3DPW. The student therefore offers lower cost with a clear accuracy loss.

The student is also 3.76 times smaller than the released baseline and has lower 3DPW PVE (110.1 versus 126.9 mm). Both the student and teacher trained on converted 3DPW images, while the released baseline did not receive this adaptation. Training the baseline on the same data would make the comparison more informative about the architecture itself.

**Video application.** The original walking run gives 356 student matches and 354 baseline matches in 356 frames. These counts show whether the tracker found a person; they do not measure pose accuracy. Later versions add temporal filtering and landmark alignment. Filtering makes consecutive outputs depend on one another, while alignment adjusts where the body is drawn. Both operate after the image prediction and require no new student training.

The inspected example motivated two separate changes: smoothing the pose and shape, then stabilizing the depth adjustment because independent alignment could make the mesh grow and shrink. A complete numerical comparison of raw, filtered and aligned outputs is still needed. The older coverage counts must not be treated as measurements of the later refinement settings.

## 6.2. Interpreting Geometry and Placement

PVE first moves the predicted and target pelvises to the same point. It measures body geometry after removing the whole-body translation. A mesh can therefore have low PVE and still appear behind or beside the person in the image. In the teacher comparison, root error changes from 1,402 to 728 mm while PVE changes from 77.4 to 73.7 mm. This is why I report placement as well as centred body error. PVE still measures differences in the body's physical size; only PA-PVE also removes a fitted uniform scale.

A whole-body average can hide an error in a small region. Two neck rotations can partly cancel and leave the head facing the right direction while twisting the skin between them. The bone-angle and neck-girth checks reveal this problem more clearly than PVE alone.

Two identical short pilots differed by 7.6 mm PVE. This warns against reading too much into a small difference between single runs. It does not provide a general error margin for every experiment. Repeated training would be needed to estimate how much each reported result varies.

## 6.3. Limitations

**Shape and children.** The teacher separates age groups better after mixed training, but still gives adult-like bodies to some children. The synthetic holdout contains no children, and 3DPW's converted age values are constant. The shape-mask experiments change both parameter variation and gender predictions, without providing a reliable fix. These findings concern the evaluated teacher variants; the student needs its own real-image shape evaluation.

**Depth and detection.** Final's predicted-versus-reference depth slope remains 0.42. Consistent anchors reduce placement error but do not solve the uncertainty of estimating distance from one image. Landmark alignment can improve an overlay while retaining a wrong body shape or pose. Its landmarks are estimates too, and their anatomical locations only approximately match the Anny joints.

**Evaluation conventions.** The 3DPW scores use fitted Anny targets and 163 Anny joints, so they are not published SMPL-space scores. Conversion error cannot be subtracted from network error. Unlabelled bystanders affect precision, and the released baseline may have seen the selected synthetic holdout during its own training.

**Experimental coverage and speed.** Each model has one recorded training run, and the student losses were not all tested separately. The walking clip has no ground-truth meshes. Natural movement and camera motion affect its motion statistics, while smoothing can hide real movement or introduce delay. The single-person example also says little about occlusion, camera cuts or crowded scenes.

Latency depends on the GPU, precision, image size and number of people. The paired model benchmark and video timings measure different workloads. In particular, alignment FPS excludes offline landmark extraction and cannot describe the speed of a complete online system.

**Reproduction.** The repository records settings and the sources of reported numbers. Some scripts still contain cluster paths, and the large data files and checkpoints are stored separately. Releasing these files requires checking the terms of the source datasets, checkpoints and body-model assets.

## 6.4. Practical Implications

For this project, I need to ask four separate questions: was the person detected, is the body correct, does it line up with the image, and is it stable across frames? Recall, reconstruction scores, placement checks and temporal measurements answer different parts of this. A useful comparison reports them alongside speed, instead of judging a model from one score or one good-looking video.

# 7. Conclusion and Future Work

## 7.1. Conclusion

In this thesis, I trained and evaluated a smaller student for multi-person human mesh recovery with Anny. The student uses a ViT-B encoder, a trainable projector and prediction heads copied from a ViT-L teacher. It learns from AnnyOne, converted 3DPW labels and the teacher's features and outputs. The teacher is only needed during training.

The 3DPW converter fits Anny bodies to source joints and keeps one body shape per person across a sequence. Virtual joints account for differences in where the two body models define their joints. The checks also show which labels remain approximate or constant. This makes the real-image results easier to interpret.

The student has 101.38 M parameters and needs 603.3 GFLOPs in the recorded configuration. Its paired L40S forward time is 28.5 ms, compared with 59.7 ms for the teacher. However, its PVE of 119.6 mm on AnnyOne and 110.1 mm on converted 3DPW does not meet the accuracy-retention target against the teacher. Its lower 3DPW error than the released baseline also reflects different training data. The result is a smaller, faster model with a measurable accuracy cost.

The video tool compares the student and baseline on the same frames. Optional temporal filtering smooths the predicted rotations, shape and position; landmark alignment adjusts placement; and a depth filter limits sudden changes in apparent size. These are inference steps around a frame-wise image model. They do not establish improved 3D accuracy without labelled video evaluation.

## 7.2. Future Work

**Reduce the student's accuracy gap.** The synthetic validation curve was still improving after 30 epochs, so a longer run is worth testing. The real-image curve had nearly flattened, which suggests that duration alone may not close that gap. Distillation weights and feature-transfer choices should be tested separately. A baseline trained on the same real images would help distinguish the benefit of the data from the benefit of the method.

**Evaluate video refinement.** The next step is to compare raw, filtered and aligned predictions on several clips, including occlusion and camera motion. The evaluation should measure stability, lag and reconstruction accuracy together. The depth filter also needs a measured comparison with independent

per-frame alignment. Learned temporal refinement can then be tested against these simpler filters. Speed measurements should include the landmark detector when discussing a complete online system.

**Improve real-image shape.** Real age-group labels, body measurements or additional Anny fits could provide supervision beyond the converter's fixed values. A held-out set with children is needed to test whether the network learns child proportions, rather than only producing them on familiar synthetic examples.

**Repeat and document the experiments.** Repeated runs and a larger evaluation set would make small differences easier to assess. The report uses the verified 30-epoch student; any later checkpoint should be reported only after its evaluation is complete. Portable configuration and clear instructions would also make the work easier to reproduce.

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# A. Reproduction: Model Names, Runs and Job Ids

Every number in this thesis can be traced in two steps. Each headline number is a macro in one of the files `numbers_*.tex` of the thesis source, and the comment next to it names the Slurm job and the line of its log, for example `3611336:191`. Table A.2 maps that job to the model, the log and the checkpoint, and Table A.1 maps every model name to the run name it had during the project. This appendix is the only place where job ids and old run names appear.

## A.1. Model names

"L" is our ViT-L line, every model that keeps the ViT-L backbone. L-stage 1 to 4 are the steps of the transfer schedule that turned NAVER's released model into ours; Base is its end and the start of every later fine-tune. Final and Control are the two arms of the last experiment, the same 10-epoch fine-tune of Base with and without the two fixes. The number after "Student-S/" is the old version number of a ViT-S student; Student is the ViT-B model distilled from Final.

**Table A.1.:** *Model names: the thesis name of every model, the run or checkpoint name it had during the project ("ep." = the epoch kept), the weights it started from, the backbone, the training data and the schedule, with the setting that is new in the row. Backbone: L, B, S = DINOv2 ViT-L, ViT-B, ViT-S with 14-pixel patches; "all" = every block trained, "last 4" = the last 4 of 24 blocks, "+p" = a linear projector to the heads' width 1024. Data: A = AnnyOne; A+P = 70 % AnnyOne and 30 % 3DPW train, with the old labels (before their correction, Chapter 3) or the corrected v23 labels. Schedule: epochs × iterations per epoch × batch, then the learning rate, halved every 30 epochs unless noted (planned epochs where the run stopped early). Checkpoints lie in `logs/anny_model/<run>/checkpoints/` with copies in `models/multiHMR/`; Table A.2 gives the path of every model with a number in the thesis, and Appendix C the full flags of Final, Control and the Student. "Not recorded": no script, log or document in the repository states it.*

| Thesis name | Old run / checkpoint name | Starting weights | Backb. | Data | Schedule; what is new |
|---|---|---|---|---|---|
| *Released starting point and baseline* | | | | | |

| Thesis name | Old run / checkpoint name | Starting weights | Backb. | Data | Schedule; what is new |
|---|---|---|---|---|---|
| Multi-HMR-Anny | `multiHMR_672_L_anny` (NAVER's release) | trained by NAVER | L, own decoder | A, BEDLAM, COCO, MPII | not recorded; evaluated with its own model class |
| *Our ViT-L line ("L"), the transfer schedule* | | | | | |
| L-stage 1 | `anny_s1_pretrained_bb_v2` | Multi-HMR-Anny backbone, fresh heads | L, all | A | $150 \times 1{,}000 \times 4$, $10^{-4}$, from epoch 177 at $3 \cdot 10^{-5}$; side branch, no checkpoint kept |
| L-stage 2 | `anny_s2_frozen_bb_v2`, ep. 299 ("v2") | Multi-HMR-Anny backbone, fresh heads | L, frozen | A | $150 \times 1{,}000 \times 4$, $10^{-4}$, from epoch 283 at $3 \cdot 10^{-5}$ |
| L-stage 3 | `anny_s2_partialft_v3`, ep. 54 ("v3") | L-stage 2, epoch 299 | L, last 4 | A | $300 \times 1{,}000 \times 4$, $10^{-5}$; helper bones held at rest |
| L-stage 4 | `anny_s2_boosted_v4`, ep. 299 ("v4") | L-stage 3, epoch 54 | L, last 4 | A | $300 \times 1{,}000 \times 4$, $10^{-5}$; neck and hand bones weighted $\times 4$ |
| Base | `anny_s2_shape_v5`, ep. 99 ("v5") | L-stage 4, epoch 299 | L, last 4 | A | $100 \times 1{,}000 \times 4$, $10^{-5}$; shape weight 20 |
| Mixed-L | `anny_teacher_v9_mixed`, ep. 62 and 149 ("v9") | Base | L, last 4 | A+P, old | $150 \times 1{,}000 \times 4$, $10^{-5}$, halved every 60 |
| Fresh-heads-L | `anny_v10_scratch_mixed`, ep. 290 ("v10", not from scratch) | Multi-HMR-Anny backbone, fresh heads | L, last 4 | A+P, old | $300 \times 1{,}000 \times 4$, $10^{-5}$; 3DPW from the first step, no neck or hand weight |
| *The last experiment: two arms and two shape-mask variants* | | | | | |
| Control | exp7 `v5_root_dist1_nofix_v23_e10`, ep. 9; job 3611339 | Base | L, last 4 | A+P, v23 | $10 \times 1{,}000 \times 4$, $10^{-5}$; 3DPW hands masked, pelvis targets |
| Final | exp7 `v5_pc_dist5_bonefix_v23_e10`, ep. 9; job 3611336 | Base and its bone correction | L, last 4 | A+P, v23 | as Control, with head targets, depth and translation weights 5, bone correction |
| Final-masked | exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask`, ep. 9; job 3612325 | as Final | L, last 4 | A+P, v23 | as Final, 8 converter constants masked for 3DPW people |
| Final-partial-mask | exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask_noage`, ep. 9; job 3614042 | as Final | L, last 4 | A+P, v23 | as Final, the 7 constants other than age masked; not adopted |
| *ViT-S students ("Student-S"), history; teacher Base unless noted* | | | | | |
| Student-S/1 | distillation v1, `anny_distill_vits` | ImageNet DINOv2-S, random heads | S, all | A | $300 \times 1{,}000 \times 4$, $10^{-4}$; feature distillation |
| Student-S/2 | v2, `anny_distill_vits_v2_a` | Student-S/1, epoch 199 | S, all | A | $200 \times 1{,}000 \times 4$, $2 \cdot 10^{-5}$, halved every 60 |
| Student-S/3 | v3, `anny_distill_vits_v3` | ImageNet DINOv2-S; 54 of 67 head tensors from Base | S, all | A | $150 \times 1{,}000 \times 4$, $5 \cdot 10^{-5}$, halved every 60; output distillation |
| Student-S/4 | v4, `anny_distill_vits_v4` | NAVER's `multiHMR_672_S` backbone; heads as /3 | S, all | A | $150 \times 1{,}000 \times 8$, $7 \cdot 10^{-5}$, halved every 60 |

| Thesis name | Old run / checkpoint name | Starting weights | Backb. | Data | Schedule; what is new |
|---|---|---|---|---|---|
| Student-S/5 | v5, `anny_distill_vits_v5` | as Student-S/4 | S, all | A | as /4; heads frozen for 5 epochs |
| Student-S/6 | v6, `anny_distill_vits_v6`, ep. 137 | NAVER's ViT-S backbone; all 67 head tensors from Base | S+p | A | as /4; stopped at epoch 138 by the time limit |
| Student-S/6b | v6b, `anny_distill_vits_v6b`, ep. 140 | Student-S/6, epoch 137 | S+p | A | 150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60; teacher in bfloat16 |
| Student-S/6c | v6c, `anny_distill_vits_v6c`, ep. 149 | Student-S/6b, epoch 140 | S+p | A | 150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60 |
| Student-S/mixed | v7, `anny_distill_vits_v7_mixed`, ep. 130 | Student-S/6b, epoch 140 | S+p | A+P, old | 150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60 |
| Student-S/final | v8, `anny_distill_vits_v8_final`, ep. 52 | Student-S/6c, epoch 149; teacher Mixed-L | S+p | A+P, old | 150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60 |
| *ViT-B student (requirement 1b)* | | | | | |
| Student | exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30`; job 3614191 (done 19 Oct) | NAVER's `multiHMR_672_B` backbone; every head tensor from Final; teacher Final | B+p, all | A+P, v23 | 30 × 1,000 × 8, $3 \cdot 10^{-5}$, halved every 12; Final's targets and masks; holdout 119.6, 3DPW 110.1 mm, 95 % rule not met |

## A.2. Environment

All training, evaluation and demo jobs ran on the DFKI Pegasus Slurm cluster, from the repository clone in `/netscratch/najib/multi-hmr`. The repository is public on GitHub (`saadnajib/Multi-Person-Whole-Body-Human-Mesh-Recovery-in-a-Single-Shot-with-Annyone-dataset-and-3dpw-dataset`); checkpoints, logs and datasets are not in it. Each job script prints the short git hash of the clone into the run's `settings.txt`, so a run can be matched to a commit.

Every job runs inside the container image `ubuntu+20.04_v4.sqsh` (in `/netscratch/najib/ContainerImages/`), started by `srun` with the option `--container-image` and the mounts `/fscratch`, `/netscratch`, `/home/najib` and `/dev/shm`. Inside it, `slurm/env.sh` is sourced first. It activates the conda environment `multihmr` (Python 3.9, PyTorch 2.0.1 with CUDA 11.7, xformers 0.0.20, torchvision 0.15.2), moves the Hugging Face, Torch and Anny caches to `/netscratch/najib`, sets one thread per data-loader worker, and strips the `X | None` type hints from the downloaded DINOv2 hub code, which Python 3.9 rejects. `slurm/egl.sh` then installs the EGL libraries for the headless renderer; the video launcher additionally tests a real render and stops before checkpoint loading if it fails.

That PyTorch build has GPU kernels up to compute capability 8.6, so the training and evaluation scripts ask for A100, A40, L40S and RTX A6000 nodes and refuse an H100 or H200 (compute capability 9.0). The V100

nodes `gifu` and `glendale` are excluded: every converter task sent there failed within a minute with an argument-parser error whose cause is unknown. Only the 3DPW converter ran on H100 nodes, through a small virtual environment with PyTorch 2.1.2 for CUDA 12.1 on top of `multihmr` (`PY_VENV=/netscratch/najib/venv_cu121`). Steps that need only NumPy (building the 3DPW label files, the re-fit check) run on the login node. The model-cost script needs a GPU node, because `model.py` imports the renderer and the login node has no EGL.

## A.3. Runs, jobs, logs and checkpoints

Table A.2 lists every job behind a number, a table or a figure of the thesis; all of them have finished.

**Table A.2.:** *The Slurm jobs behind every number, table and figure of the thesis, with the log and the checkpoint or output. Model names as in Table A.1. Paths are relative to the project folder on the cluster; the log of job j is `logs/<logname>_j.out` (array jobs: `_j_<task>.out`). `E7` stands for `results/experiments/exp7_final_model/`, `M` for `models/multiHMR/`.*

| Model or item: what the job did | Job(s) | Log name; checkpoint or output |
|---|---|---|
| *Models* | | |
| Base: holdout trace (V1 PVE, placement) | 3570222; re-trace 3571596 | `diag_holdout`; `logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt` |
| Mixed-L: holdout trace | 3571963 | `diag_holdout`; `M/anny_teacher_v9_mixed_00062.pt` |
| Fresh-heads-L: holdout trace | 3571794 | `diag_holdout`; `logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt` |
| Final: training, holdout trace, 3DPW test | 3611336 (smoke test 3611333) | `exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt`, copy `M/exp7_v5_final_00009.pt` |
| Control: the same | 3611339 | `exp7_final_model`; `E7/v5_root_dist1_nofix_v23_e10/` |
| Final-masked: the same | 3612325 | `exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10_shpmask/` |
| Final-partial-mask: the same (done 19 Oct) | 3614042 | `exp7_final_model` (holdout table lines 191–215); `E7/v5_pc_dist5_bonefix_v23_e10_shpmask_noage/` |
| Student: distillation from Final, 30 epochs, own evaluation (done 19 Oct) | 3614191 (smoke test 3613926) | `exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt`; `final_table.json` in the same run folder |
| Multi-HMR-Anny: holdout and 3DPW test with our code | 3614192 (3613928 crashed: FFN width) | `eval-test3dpw` (task 0); `M/multiHMR_672_L_anny.pt` |
| Model cost: parameters, GFLOPs, latency of Multi-HMR-Anny and Final (A100) | 3614947 | `model_cost`; `results/diagnostics/model_cost_naver_vs_final.json` |
| *Evaluations and demos* | | |
| 3DPW test (v23): Base, Mixed-L, neck pilot (Final and Control in their own jobs) | 3612073, 3612074, 3612075 | `eval-test3dpw` (task 0), lines 117–125 |

| Model or item: what the job did | Job(s) | Log name; checkpoint or output |
|---|---|---|
| 3DPW test (v23): Student-S/6c, Student-S/final, also AnnyOne test-100 | 3615020_2, 3615020_4 | `eval-test3dpw`, line 119 (_2) and 118 (_4) |
| Relative Human: Base, neck pilot, Final, Final-masked, Final-partial-mask | 3611285_0, _1, 3612131, 3613616, 3616821 | `eval_rh`; `results/diagnostics/relative_human_val_<tag>/` |
| Demos: neck pilot, Final, Final-masked, each next to Base | 3581312, 3612078, 3613617 | `demo-compare`; `demo_montage_photos/`, `demo_montage_3dpwtest/` |
| Demos: predicted fingers (Base, Final); rest pose (Base, Final, Final-masked) | 3615018, 3615019 | `demo-compare`; scripts `slurm/demo/run_demo_compare_hands.sh`, `_rest.sh` (copies made on the cluster) |
| *3DPW labels* | | |
| Label check before and after the re-fit | 3575369, 3611284 (v23) | `check_3dpw_conversion`; `results/diagnostics/check_3dpw_conversion_v23/` |
| Re-fit with converter v2.3, 87 person-sequences | 3590385, 3602813 | `smpl2anny-all`; `threedpw_anny_v23/labels/`; `logs/check_refit_all_v23.log` |
| Label rebuilds from the old fits | 3579266, 3581320 | `3dpw-labels`; `threedpw_anny_full/labels_rebuilt/` |
| *Pilots and diagnostics (outputs in `results/experiments/` and `results/diagnostics/`)* | | |
| Phenotypes; youngest 50 training people | 3569948; 3569950 | `diag_pheno`; `diag_young` |
| Real-photo ages with the crop camera: Base, sampler pilot, its control | 3569949, 3572419, 3572420 | `diag_cropK` |
| Depth pilots, 2 epochs (×1, control, ×5, ×20) and 10 epochs (×1, control, ×5) | 3571598, 3571599, 3572418, 3572483; 3571933, 3571934, 3572482 | `exp1_depth_anchor` |
| Sampler pilot and control; batch ages only | 3571955, 3571956; 3571601 | `exp2_children` |
| Head-loss pilot; two identical controls (noise) | 3571957; 3571958, 3571960 | `exp3_neck_global_rot`; `exp4_fingers` (3571960) |
| Finger pilot (control 3571960: crop trace, hand size) | 3571959 | `exp4_fingers` |
| Geodesic pilot and its L1 control | 3573520, 3573521 | `exp5_neck_fix` |
| Per-bone dumps of eight checkpoints | 3573518, 3573519 | `diag_neck_bones`; `results/diagnostics/neck_bones/` |
| Bone correction: Base (with the neck pilot), Fresh-heads-L | 3579014, 3579015 | `exp6_bone_fix`; `results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json`; neck pilot `M/exp6_v5_bonefix_00001.pt` |

The scripts behind these jobs and the exact commands that produced the thesis numbers are listed in the file `docs/thesis/REPRODUCE.md` of the repository. Each command runs from the login node in the repository clone; the flags each training run passes are in Appendix C.

## A.4. What cannot be reproduced exactly

The training code sets no random seed. The `-seed` of the holdout trace only picks which training images the bone correction is estimated on. The order in which training images are drawn (with replacement) therefore differs between

two runs of the same command, and so does the number of optimiser steps skipped for non-finite gradients (16 in Final, 10 in Control, out of 10,000). Two runs of one 2-epoch configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7); these two pilots do not establish an expected interval for a repeat of Final or Control. The two arms also ran on different GPU types (Final on an A100, Control on an A40, both with TF32), which changes the speed and, at the level of rounding, the arithmetic. Evaluating a fixed checkpoint involves no sampling; that it repeats exactly across GPU types was not tested.

Three further limits apply. The history scripts of the L line and of Student-S were edited after their runs, so their recipes in Table A.1 are as recorded there, not re-run. `train.py` keeps only the last two or three checkpoints of a run, so intermediate epochs are gone. The checkpoints, the AnnyOne and 3DPW data and the Relative Human annotations are not in the repository; 3DPW and Relative Human come with their own licences.

## A.5. Video Execution Evidence

The original walking run is recorded in `video-compare-3642415.out` and its corresponding error log. The run used the 30-epoch student checkpoint `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt` and the correction selected by that run's `student_fix_arg.txt`. The baseline was `models/multiHMR/multiHMR_672_L_anny.pt`; the input was `man_walking.mp4`. Both checkpoints loaded with 100% state coverage. The renderer reported an NVIDIA L40S and passed its test mesh.

The final log lines report 356 processed frames for each model, 356 student matches and 354 baseline matches. The completed output is `results/video_compare/video_3642415/comparison.mp4`. The log contains no synchronized inference-FPS result from the later timing implementation and supplies no numerical pose/shape stability summary.

The video implementation and EGL correction are recorded in commits `1d5621b` and `7b0b5e1`; the revised FPS implementation is `a0fa59f`. It is launched through `slurm/demo/run_video_compare.sh`. Its usage, timing boundaries and outputs are documented in `docs/video_inference.md`.

Later video revisions add causal parameter filtering (`9a7a1d5`), landmark alignment (`ef7759f`) and aligned-depth filtering (`5e70080`). The current method is described in Section 4.10. The raw-run counts above predate these options. The later comparison video supplied during review showed apparent size changes after alignment; this motivated the depth filter, but no new ground-truth video metrics were supplied.

# B. SMPL–Anny Joint Correspondence

The converter compares the rest skeletons geometrically before fitting posed frames. Hip targets are associated with Anny's proximal upper-leg bones by proximity to the hip midpoint. For the spine, candidate Anny bones are ordered by normalized height and the ordered triple nearest the SMPL spine heights is selected. Limb and head candidates start from the anatomical correspondence in the converter. The final per-run choices are saved with the fits.

Let $\mathbf{o}_A, \mathbf{o}_S$ be hip midpoints, $s$ the ratio of hip-to-neck lengths and $\mathbf{Q}_{S\to A}$ the rest-frame axis alignment. The target location expressed in Anny's rest frame is

$$\mathbf{j}_k^{S\to A} = \mathbf{o}_A + s\mathbf{Q}_{S\to A}(\mathbf{j}_k^S - \mathbf{o}_S), \qquad \boldsymbol{\delta}_k = (\mathbf{Q}_b^0)^\top(\mathbf{j}_k^{S\to A} - \mathbf{h}_b^0), \tag{B.1}$$

for the associated bone $b$. The posed virtual joint then follows Equation (3.2). The offset is fixed after rest-pose construction; it is not an extra joint learned independently in each frame.

| SMPL target | Anny correspondence and fitting point |
|---|---|
| Pelvis | root bone with a measured torso offset |
| Left/right hip | proximal upper-leg bone selected geometrically, with an offset |
| Spine1, spine2, spine3 | ordered rest-height correspondence among spine bones, with offsets |
| Neck | neck candidate with a measured rest-frame offset |
| Left/right collar | clavicle bone with an offset |
| Shoulder, elbow, wrist | corresponding upper-arm, lower-arm and wrist bone heads |
| Knee, ankle, foot | corresponding lower-leg, foot and toe bone heads |
| Head and hand targets | corresponding candidate bone heads; pose determined by the permitted parent rotations |

**Table B.1.:** *Correspondence construction used by the converter. Exact resolved bone names and offset vectors are metadata of the fit, rather than universal constants shared by every Anny rig version. The fit rotates the root and 18 mapped body bones; remaining bones, including most fingers and fine head/neck articulation, retain their rest rotation.*

The fitting output stores `anny_bone_names`, `joint_offsets_local` and `offset_joints`. These arrays are the authoritative correspondence for reproducing a particular fit. The code-level construction is in `scripts/data/smpl_`

`to_anny.py`; the effect of geometric matching and virtual joints is evaluated in Appendix G.

# C. Flags and Weights

This appendix lists the command-line flags of the training runs of Final and Control (Table C.1) and of the Student (Table C.2) that differ from the default in the code, or that matter although they keep it; every other flag keeps its default. The loss weights are in Table 4.1 of Chapter 4, the commands that start the runs in the file `docs/thesis/REPRODUCE.md` of the repository (Appendix A).

**Table C.1.:** *Training flags of Final and Control. "Value" is Final's; where Control differs, its value follows in brackets. Paths are relative to the project folder on the cluster. Both runs start from Base and see 10 epochs of 1,000 iterations at batch 4, that is 40,000 sampled image presentations, at a constant learning rate.*

| Flag | Default | Value (Control) | Meaning |
|---|---|---|---|
| *Start and network* | | | |
| `--pretrained` | none | Base checkpoint | weights the run starts from, heads included |
| `--pretrained_remap` | 1 | 0 | no renaming of weight names; Base was saved by this code |
| `--backbone`, `--img_size` | ViT-S, 336 | `dinov2_vitl14`, 672 | DINOv2 ViT-L, 48 × 48 tokens of 14 pixels |
| `--freeze_backbone`, `--unfreeze_last_n_blocks` | 0, 0 | 1, 4 | backbone frozen except its last 4 of 24 blocks and final norm |
| `--bone_fix` | off | Base's `bone_fix.json` (`none`) | constant output rotation on neck02, head, metacarpal1.L and finger4-1.R, stored in the checkpoint (Section 4.6) |
| *Targets and losses* | | | |
| `--gt_center_joint` | root | `person_center` (`root`) | the joint the depth, translation, cell and offset targets describe: the head, which the network anchors (Control: the pelvis) |
| `--alpha_dist`, `--alpha_transl` | 1, 1 | 5, 5 (1, 1) | weights of the depth and translation terms |
| `--alpha_shape`, `--alpha_j2d`, `--alpha_v2d` | 1, 1, 1 | 20, 0.1, 0.1 | shape and 2D loss weights |
| `--boost_neck_weight`, `--boost_hand_weight` | 1, 1 | 4, 4 | extra weight on neck, head and hand bones, renormalised to mean 1 |
| `--start_2d_epoch` | 10 | 0 | 2D terms from the first epoch |
| `--mask_helper_joints_train` | 0 | 1 | hold the 88 helper bones at identity, no rotation targets |
| `--use_anny_shape` | 0 | 1 | phenotype targets from the dataset |
| `--mask_3dpw_hands` | 0 | 1 | no rotation targets for the 38 hand bones of 3DPW people |
| `--mask_3dpw_shape` | 0 | not passed | shape mask for 3DPW people; 1 only in Final-masked and Final-partial-mask |
| *Data* | | | |

| Flag | Default | Value (Control) | Meaning |
|---|---|---|---|
| `--train_data`, `--val_anny_n` | BEDLAM, 0 | AnnyOne, 100 | AnnyOne for training, its last 100 images as the holdout |
| `--exclude_indices_file` | none | the 55 manifest images | never sample the diagnostic-manifest images |
| `--mix_3dpw_labels` | none | `threedpw_anny_v23/labels` | corrected 3DPW labels mixed in, 30 % of a batch on average (default ratio 0.3) |
| `--val_3dpw_n` | 0 | 300 | evaluate on 300 evenly spaced 3DPW validation images |
| *Optimisation, bookkeeping, job* | | | |
| `--batch_size`, `--learning_rate` | 2, $5 \cdot 10^{-6}$ | 4, $10^{-5}$ | images per step; Adam learning rate |
| `--n_iters_per_epoch`, `--max_iter` | 100, 10,000 | 1,000, 10,000 | 10 epochs of 1,000 steps |
| `--amp` | 1 | 0 | no mixed precision (it produced NaNs with the Anny layer); TF32 stays on |
| `--num_workers`, `--nb_max_ckpt`, `--log_freq` | 0, 10, 100 | 12, 2, 50 | bookkeeping; output in `results/experiments/exp7_final_model/<run>/train` |
| `TIME`, `EVAL_TEST` (script) | | 5 h, 1 | Slurm limit on A100, L40S, RTX A6000; after training the holdout trace and the 3DPW test (v23) |

**Table C.2.:** *Training flags of the Student where they differ from Final's. Every other flag, the targets, masks, loss weights and 3DPW mixing included, is as in Table C.1, so that the Student learns the corrected depth; no shape mask. 30 epochs of 1,000 iterations at batch 8, that is 240,000 sampled image presentations, with a 20-hour Slurm limit. The colour-jitter flags (`-brightness 0.2` and so on) are passed but read nowhere in the code, so no augmentation was used.*

| Flag | Default | Value | Meaning |
|---|---|---|---|
| `--backbone` | ViT-S | `dinov2_vitb14` | DINOv2 ViT-B |
| `--pretrained` | none | `models/multiHMR/multiHMR_672_B.pt` | NAVER's ViT-B as the backbone start |
| `--pretrained_remap`, `--load_only_backbone`, `--freeze_backbone` | 1, 0, 0 | 1, 1, 0 | rename NAVER's weight names; load the backbone only and train all of it |
| `--head_dim` | 0 | 1024 | heads at the teacher's width, fed by a 768 to 1024 projector |
| `--distill_teacher_ckpt`, `--teacher_bf16` | none, 0 | Final, 1 | frozen teacher, its backbone in bfloat16; turns the run into distillation |
| `--init_heads_from_teacher`, `--freeze_heads_epochs` | 0, 0 | 1, 5 | copy every matching head tensor from the teacher; keep them frozen for 5 epochs |
| `--lambda_kd`, `--lambda_kd_out` | 1.0, 0 | 1.0, 1.0 | feature distillation (KL, temperature 4) and output distillation (L1) |
| `--bone_fix` | off | the teacher's | Final's correction in training; the Student's own, estimated on 200 training images, for evaluation |
| `--batch_size`, `--learning_rate` | 2, $5 \cdot 10^{-6}$ | 8, $3 \cdot 10^{-5}$ | images per step; Adam learning rate |
| `--max_iter`, `--lr_decay_every` | 10,000, 30 | 30,000, 12 | 30 epochs; learning rate halved after epochs 12 and 24 |
| `--eval_freq`, `--nb_max_ckpt` | 1, 10 | 5, 3 | evaluate every 5 epochs, keep 3 checkpoints |

# D. Extra Figures

This appendix holds the figures that the chapters refer to but do not show: the second depth example, two more bone-correction pairs, a second neck pair on 3DPW, the other demo frames and photos, all twelve demo images with predicted fingers and nine more rest-pose line-ups. Every panel is judged by eye; where a body is placed from the detector box, depth errors are hidden.




**Figure D.1.:** *Holdout image 579860 rendered at each model's own predicted depth: ground truth, Base, and the depth pilot (head-joint targets, depth weight ×5, 10 epochs, AnnyOne only). Before the fix the ground-truth head shows above a shrunken body placed too far away. Synthetic image; one case, not a typical one. Companion of Figure 5.5.*

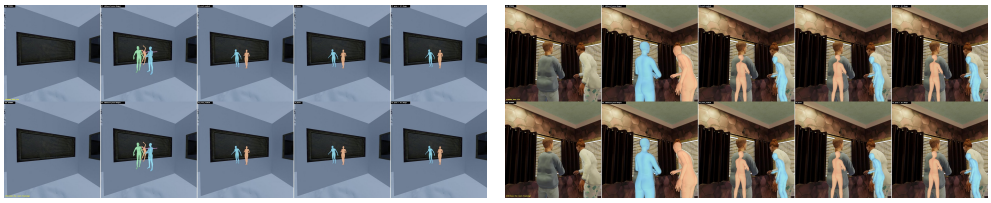


**Figure D.2.:** *Holdout images 579864 (left) and 579866 (right): Base as trained next to the same checkpoint with the bone correction applied at the output, no training (Section F.4). Synthetic images at the model's own predicted depth, so the depth bug still draws the bodies too small; necks judged by eye. Companions of Figure F.1.*

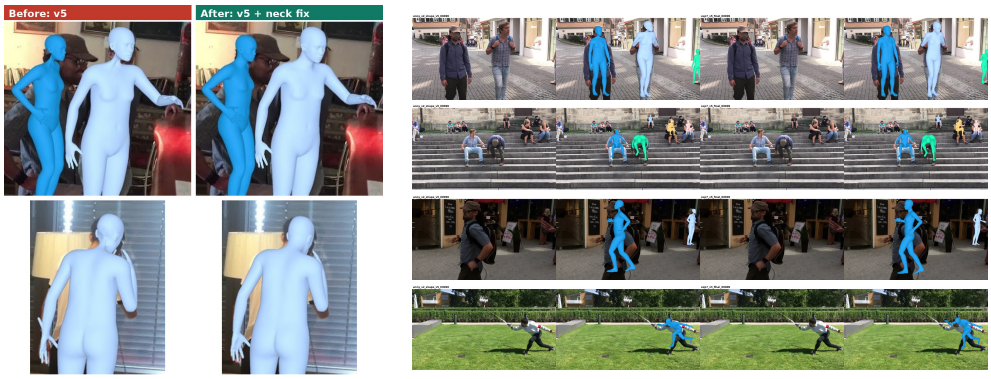



**Figure D.3.:** *Left: 3DPW test frames* `downtown_bar_00` *701 (top) and* `office_phoneCall_00` *440 (bottom), Base next to the neck pilot (the bone correction and a 2-epoch fine-tune); necks ringed. Bodies placed from the detector box, fingers at rest pose; judged by eye, no metric. The two columns differ by the correction and by the fine-tune.*

**Figure D.4.:** *Right: the other four 3DPW test frames of the demo, top to bottom* `downtown_walking`*, stairs,* `downtown_runForBus` *and fencing, each as photo and Base, then photo and Final (labels in the image). Placed from the detector box, fingers at rest pose; by eye, no metric. The first four frames are in Figure 5.4.*



**Figure D.5.:** *The four real photos of the demo (garden, tennis; stage, party), never used in training, each as photo and Base, then photo and Final; placed from the detector box, fingers at rest pose; by eye, no metric. The tennis photo also appears in Figure 5.10.*

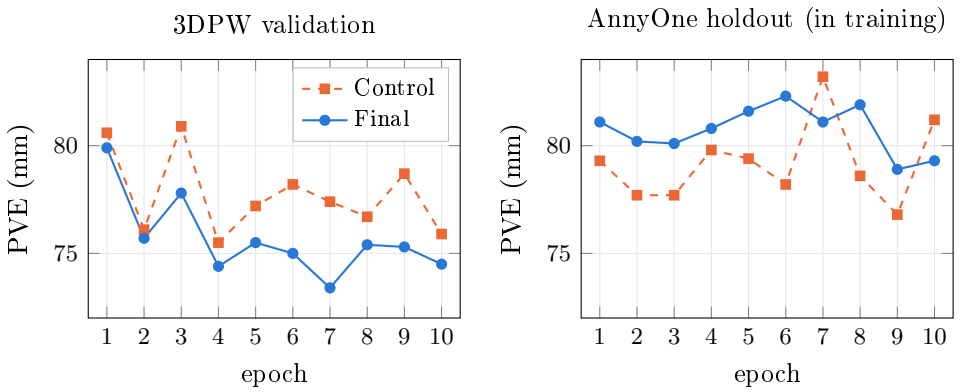



**Figure D.6.:** *PVE per epoch during training, Final and Control (one run each, no seed). Left: 300 images of the corrected 3DPW validation labels. Right: the AnnyOne holdout as evaluated during training, setting V0 (helper bones unmasked), so higher than the V1 numbers of Table 5.5a. Values from the training logs; read in Section 5.5.1.*



**Figure D.7.:** *All twelve demo images with predicted fingers (not set to rest pose), each as photo and Base, then photo and Final (labels in the image). Row by row: the garden and tennis photos; the stage photo and fencing; stairs and* `downtown_walking`*;* `downtown_runForBus`*, the party photo and guitar;* `downtown_bar`*,* `office_phoneCall` *and the weekly market. Demo setting, placed from the detector box; Base still has the flipped left index-finger base (metacarpal1.L) and right ring-finger base, Final has the correction. In most images the hands are a few pixels wide and cannot be judged; Figure 5.8 zooms on the three where they can. By eye; no metric.*



**Figure D.8.:** *The other nine demo images in rest pose, each as photo and Base, photo and Final, photo and Final-masked (labels in the image). Top to bottom: the garden photo, the 3DPW test frames fencing, stairs,* `downtown_walking` *and* `downtown_runForBus`*; then the party photo and guitar,* `office_phoneCall` *and the weekly market. Every detected person is in rest pose, so only body shape differs. Demo setting, placed from the detector box; at this size only gross differences of build can be seen. By eye; no metric. The other three images are in Figure 5.11.*

# E. Pilot Results in Full

Before the final runs, six short pilots tested one change each against a control arm. Table E.1 gives, for each, what it changed, the success rule written into its script before it ran, the result and the control. All pilots fine-tune Base on AnnyOne only, at learning rate $10^{-5}$ and batch 4, for 2 epochs (8,000 images) unless noted, and are read on the 100-image AnnyOne holdout in setting V1. Each is one run without a seed, so every result must be read against the run-to-run variation of Section F.7, summarised below the table. Appendix F explains what each result means; the full depth runs are Table F.3.

**Table E.1.:** *The six pilot experiments. V1, 100-image AnnyOne holdout, matched people; fine-tunes of Base, one run each, without a seed. The rule is the success rule written into each script before the run. Head direction is the global head orientation error.*

| Pilot | What changed | Success rule | Result | Control |
|---|---|---|---|---|
| Depth pilots (2 and 10 epochs) | training targets at the head joint instead of the pelvis; depth and translation weights ×1, ×5, ×20 | slope toward 1 and intercept toward 0 m, with PVE ≤ 71.5 mm and recall ≥ 89.2 % | rule not met (PVE or recall missed in every run, controls too); root error 1.21 → 0.54 m, size 0.63 → 0.84, slope 0.33 → 0.38 (×5, 10 epochs) | same schedule, targets at the pelvis, weight ×1 (one 2-epoch, one 10-epoch) |
| Sampler pilot | age-balanced sampling: children (age < 0.3) per batch 2.1 → 9.7–11.1 % | 10 manifest children: PVE < 47.8 mm and \|age bias\| < 0.078; holdout PVE ≤ 71.5, recall ≥ 89.2 | inconclusive: children 48.5 vs. 53.1 mm, age bias 0.078 vs. 0.085; baby age floor 0.100 → 0.102 | same run with the sampler off (age ×3 and height ×2 shape weights in both) |
| Head-loss pilot | L1 loss on the camera-frame orientation of head, neck03 and both wrists, weight 5 | head direction < 12.0 deg; PVE ≤ 71.5 mm, recall ≥ 89.2 % | head direction 11.3 deg, within the controls' 11.7–13.6; safety limit missed (71.8 mm, 89.0 %) | two identical runs at weight 0 |
| Finger pilot | region-normalised rotation loss (body, hands, head-neck averaged separately); a test-time person crop | fingers below the rest-pose floor 15.1 deg without a PVE cost | no effect: fingers 17.5 vs. 17.4 deg; crop 17.2 deg at body PVE 124.8 mm | same run with the loss off (one of the two identical runs above) |

| Pilot | What changed | Success rule | Result | Control |
|---|---|---|---|---|
| Geodesic pilot | geodesic angle instead of L1 on the rotation matrices | flagged bones down by $\geq 3$ deg more than the control, or neck and head under 15 deg | no movement: neck02 175.6 $\rightarrow$ 175.0, head 177.9 $\rightarrow$ 177.1 deg | same run with the L1 loss |
| Bone-correction pilot | constant output rotation on neck02, head, metacarpal1.L, finger4-1.R; no training, then a 2-epoch fine-tune | bones $< 15$ deg, girth $\geq 0.9$; no training: head direction $\leq$ Base $+ 3$ deg; fine-tuned: guards against the control | no training: fails (head direction 19.1 vs. limit 15.0 deg); fine-tuned: passes (neck02 4.7 deg, girth 0.997, fingers 8.3 deg) | the geodesic pilot's L1 control |

**Pilot variation.** Two runs of one short configuration differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7). Restarting fine-tuning also raised PVE relative to Base in the recorded runs. These observations make small single-run differences difficult to interpret; they do not define a universal significance threshold. I selected the placement and bone corrections using their targeted diagnostics as well as the body-error tables.

## E.1. History: the ViT-S students and the schedule to Base

Table E.2 keeps the draft's V0 measurements from before both fixes: the schedule to Base (Table E.2a) and the ViT-S students (Table E.2b), at best 1.54 times their teacher's error. These trials informed the final student design. Mesh-recovery pretraining and heads at the teacher's feature width performed better in the recorded trials, so Student uses NAVER's ViT-B initialization and Final's heads. I also increased the training budget to six times as many sampled image presentations. These trials do not prove that data, rather than capacity, was the only limitation.

| Run | Change against the previous run | PVE | PA-PVE | MPJPE | Rec. |
|---|---|---|---|---|---|
| L-stage 1 | fresh heads, all blocks trained (side branch) | – | – | – | – |
| L-stage 2 | fresh heads, backbone frozen | 145 | – | – | – |
| L-stage 3 | + last 4 of 24 blocks, helper mask | 73.6 | – | – | – |
| L-stage 4 | + neck and hand loss ×4 | 73.6 | 56.5 | 73.2 | – |
| Base | + shape loss weight 20 | 74.3 | 57.5 | 74.4 | 90 |

**(a)** *The transfer schedule that produced Base (Table 4.2); recall (Rec.) in %. L-stages 2–4: evaluation during training; Base: the standalone script.*

| | | | AnnyOne holdout, V0 | | | | | 3DPW test, v23[c] | |
|---|---|---|---|---|---|---|---|---|---|
| Model | Teacher | Params | PVE | PA-PVE | MPJPE | Prec. | Rec. | PVE | PA-PVE |
| Base (ViT-L) | – | 318.4 | 74.3 | 57.5 | 74.4 | 81 | 90 | 121.8 | 72.0 |
| Mixed-L | – | 318.4 | 78.5 | 57.4 | 78.6 | 83 | 89 | 95.5 | 62.9 |
| Student-S/2 | Base | 32.8 | 193.4 | 128.2 | 210.7 | – | 86 | – | – |
| Student-S/6b[a] | Base | 37 | 126.9 | 92.1 | 132.0 | 77 | 88 | – | – |
| Student-S/6c | Base | 37 | **114.5** | **84.6** | **118.4** | 81 | 89 | 142.6 | 91.0 |
| Student-S/mixed[b] | Base | 37 | 120.6 | 88.2 | 125.2 | 78 | 87 | – | – |
| Student-S/final[b] | Mixed-L | 37 | 120.6 | 86.9 | 125.0 | 76 | 87 | 121.6 | 83.7 |

**(b)** *The ViT-S students against their ViT-L teachers; parameters in millions, precision (Prec.) and recall (Rec.) in %. Student-S/2: feature distillation only; Student-S/6b: heads fed through a projector at the teacher's width; Student-S/6c: Student-S/6b at batch 16; Student-S/mixed and Student-S/final: continued with 3DPW images. [a]Evaluation during training. [b]30 % 3DPW images per batch, old labels. [c]Corrected labels, the script of Table 5.5b; Final: 89.0 / 57.6 mm.*

**Table E.2.:** *History, before the depth fix, the bone correction and the corrected 3DPW labels. AnnyOne holdout (100 images), setting V0, errors in mm; one run per model, no seed. "–": not measured with the same script. The students' old-label 3DPW numbers are retired.*

# F. Supporting Diagnostic Experiments

Base showed three failures in its renders: thin, twisted necks, claw-like fingers and adult bodies on children. I used targeted checks to investigate the causes before choosing training changes. Unless stated otherwise, every number is from Base on the 100-image AnnyOne holdout in setting V1, one run per arm without a seed, read against the variation of Section F.7.

## F.1. The three visible failures

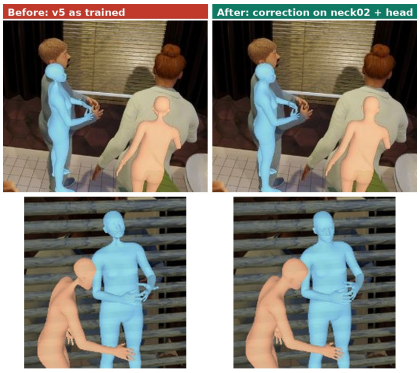



**Figure F.1.:** *Holdout images 579862 (top) and 579863 (bottom). Left: Base as trained. Right: the same checkpoint with the output correction of Section F.4, no training. Synthetic, bodies at the model's own depth, so the depth bug of Section F.3 still draws them too small. Necks judged by eye.*

In Figure F.1 (left) Base draws each neck as a thin stalk pinched under the head. The fingers curled into claws, so every real-photo demo set them to rest pose. A standing boy in one of our photos received an adult body (Figure 5.10).

The draft gave each failure an untested cause: skinning of the stacked neck bones ("a property of the rig"), finger resolution, and unlearned age, shown by five small people in our photos predicted at adult ages (0.57–0.67). The neck cause is wrong, resolution explains little, and the child evidence was wrong: four of the five were adults who crouched, sat or were cut off by the frame (height phenotype 0.47–0.56). Only the boy was a child, and he received age 0.60 and height 0.48: an adult. The number behind the necks of the figure is the girth of Table F.4.

## F.2. What is not the cause: six structural checks

| check | sample | result | verdict |
|---|---|---|---|
| 1 labels decode onto the people (no network) | manifest, 55 people | 84.5 % of joints inside the image, 84.9 % inside the box (the rest is truncation); dataset camera right on 100 % of images | labels fine |
| 2 body model, age sweep | body model | skeletal height 0.53 m at age 0, 1.55 m at age 0.8; head/height 0.200 to 0.106 | child shape exists |
| 3 single-joint moves | body model | all 16 pass (96–100 % of moved vertices in the intended part) | rig fine |
| 4 gradients | 4 holdout images | zero gradient only for the 88 helpers; all 75 active bones and 11 shape outputs get gradient (598 at the root to 4.7 at the fingers); two of three camera outputs get none | path fine |
| 5 prediction vs. truth, shape swap | holdout, manifest | neck length 99.3 % of truth; true shape lowers the 10 manifest children's PVE 47.8 → 41.6 mm | shape head is the gap |
| 6 capped overfit | 40 manifest training images | training PVE −36 % (80.7 → 51.8 mm), holdout +45 % | learns; two warnings |

**Table F.1.:** *Six checks on Base before any training change. The manifest holds 40 training images (10 adults, 10 children, 10 with visible hands, 10 hard neck poses) and 15 holdout images without children; it is excluded from every later fine-tune. Check 6 trains a throw-away copy for 400 steps and reports V0.*



**(a)** *holdout 579888, a close-up with hands* **(b)** *training image 462901, low camera*

**Figure F.2.:** *Check 1 on two manifest images: image | ground-truth mesh | joints and box, decoded through the training body model and drawn with the dataset camera; no network. By eye the mesh lies on the person, down to the fingers. In (a) the frame cuts head and legs: truncation, not a decoding error (the black corner is in the dataset image). Synthetic; two of four rendered cases.*

The sampled checks did not reveal a general failure in the rig, label decoding or gradient path. Shape dominates the loss (25.7 %; translation 1.9 %, distance 0.6 %). I therefore examined the targets and per-bone predictions more closely. The overfit run gave two warnings: the rotation loss barely moved while the images were memorised (44.6 → 40.8), and the translation and 2D terms rose (1.67 → 1.89, 58 → 75, first against last five epochs), possibly a first sign of the depth bug. That evidence is weak: the values are single-batch prints, and

the half-run averages fall (2.10 → 1.96, 69 → 63).

The checks are blind to two faults, and both were present: a target at another joint than the anchor (they compare pelvis-centred quantities) and a bone flipped the same way in every person (bone errors were averaged by group).

## F.3. The hidden depth bug

| quantity | truth | Base |
|---|---|---|
| pelvis depth, mean (sd) | 1.92 m (0.49) | 3.00 m (0.32) |
| people placed farther than truth | – | 100 % |
| corr. of true depth with depth error | – | −0.76 |
| skeletal height (3D) | 1.409 m | 1.405 m ($r = 0.89$) |
| head / pelvis 2D error, median | 0 | 46 / 14 px |
| corr. of PVE with depth error | – | −0.03 |

**Table F.2.:** *Where Base places people. V1, 100-image AnnyOne holdout, all 222 matched people; slope, intercept and drawn size are in Table F.3. Right metric size, wrong distance; PVE does not see it.*

| run (depth loss) | root (m) | size | slope | icpt. (m) | head px | pelvis px | PVE | recall |
|---|---|---|---|---|---|---|---|---|
| Base | 1.21 | 0.63 | 0.33 | 2.36 | 64.5 | 23.4 | 70.5 | 90.2 |
| *2-epoch fine-tunes of Base, AnnyOne only* | | | | | | | | |
| control ×1 | 1.11 | 0.66 | 0.33 | 2.28 | 58.1 | 23.3 | 72.3 | 89.8 |
| fix ×1 | 0.69 | 0.78 | 0.30 | 1.93 | 9.6 | 37.3 | 77.3 | 87.8 |
| fix ×5 | 0.66 | 0.80 | 0.44 | 1.62 | 8.5 | 38.1 | 76.8 | 87.8 |
| fix ×20 | 0.53 | 0.87 | 0.43 | 1.44 | 8.8 | 31.4 | 75.7 | 87.4 |
| *10-epoch fine-tunes of Base, AnnyOne only* | | | | | | | | |
| control ×1 | 1.44 | 0.59 | 0.34 | 2.59 | 64.0 | 24.3 | 72.2 | 89.4 |
| fix ×1 | 0.73 | 0.78 | 0.30 | 1.98 | 7.6 | 38.5 | 75.5 | 87.0 |
| fix ×5 | 0.54 | 0.84 | 0.38 | 1.59 | 7.2 | 33.1 | 74.8 | 87.8 |
| *10-epoch fine-tunes of Base with 30% 3DPW (v23 labels)* | | | | | | | | |
| Control ×1 | 1.40 | 0.59 | 0.28 | 2.66 | 67.1 | 26.7 | 77.4 | 89.0 |
| Final ×5 | 0.73 | 0.78 | 0.42 | 1.75 | 7.6 | 41.1 | 73.7 | 87.0 |
| *other checkpoints trained with pelvis targets* | | | | | | | | |
| Mixed-L | 1.39 | 0.59 | 0.31 | 2.60 | – | 25.1 | 75.0 | 88.6 |
| Fresh-heads-L | 1.49 | 0.57 | 0.28 | 2.71 | – | 24.0 | 74.1 | 89.0 |

**Table F.3.:** *Placement before and after building the training targets at the head joint. V1, 100-image AnnyOne holdout, matched people; metrics as in Table 4.4, head and pelvis px = 2D error, PVE in mm, recall in %. Control rows build the targets at the pelvis, fix rows at the head; ×1/5/20: distance and translation weight. Final also has the bone correction. One run per row, no seed; each fix row is read against the control of the same length (none was run at ×5 or ×20).*

Base places every person too far (Table F.2, Figure 5.6), the nearest farthest. The skeleton has the right height, so the body is drawn at 0.63 of its true size.

PVE is blind to all of it.

The code gave the cause: the mesh is placed by its head joint, but the anchor targets were built at the pelvis (Section 4.5). A one-batch debug print showed offsets of $-2$ to $-19$ cells where the legal range is $\pm 0.5$. The likely mechanism, not tested on its own: with the mesh hanging from a misplaced head, the 2D terms can shrink the misfit only by drawing the body smaller, that is, farther away, and the direct depth terms (2.5 % of the loss) could not resist; near people have the largest misfit.

Table F.3 tests the fix against same-schedule controls. All five fix runs reach a root error of 0.53–0.73 m, both controls stay at 1.11–1.44 m, the drawn size rises from 0.59–0.66 to 0.78–0.87, and the head lands within 7–10 instead of 58–67 pixels: every tested fix run has lower root error than either control. This is consistent evidence across these configurations, although repeated-run uncertainty was not estimated. Two other checkpoints trained with pelvis targets carry the bug too. It stayed hidden because PVE aligns the pelvis first and the real-photo demos place each body from its detection box.

Depth remains inaccurate: the slope barely moved (0.30 with the switch alone, 0.38–0.44 with a stronger depth loss), and predictions are still pulled toward 2.5–2.9 m for a reason I have not found. The pelvis 2D error rose to 31–41 pixels, likely because the head is now pinned while the body is still too small (not checked). Every fix run is worse than its control by $+2.6$ to $+4.9$ mm PVE and $-1.6$ to $-2.4$ recall points. The direction is consistent across these runs, but the size of any repeatable cost is not established. By the success rule written beforehand (PVE $\leq$ 71.5 mm, recall $\geq$ 89.2 %) every run failed, the controls included.

## F.4. The neck: a per-bone dump

A per-bone dump of the neck chain found the fault (Table F.4, first column). neck02 is 175.6 deg wrong, a half-turn about its own axis (twist 175.7, swing 3.3 deg), and the head bone 177.9 deg; neck03 is right relative to its parent (5.7 deg) but 173.3 deg off in the camera frame. The half-turns cancel, so the head points the right way (12.0 deg); the skin between is twisted thin. Group means (62.7 and 89.5 deg) had hidden the flip. It is in all eight checkpoints of the Base line (neck02 175.0–176.0 deg, girth 0.46–0.47), from the first stage on, before the helper mask, the neck boost and the shape weight of 20; the depth pilot keeps it, so depth is not the neck's cause. Fresh-heads-L has a normal neck (girth 1.00) but flips other bones: a compensated pair in the left arm and three metacarpals.

This refutes the draft: the rig passes all 16 single-joint moves, and rest pose (all rotations identity) hides the flip. A loss did not undo it: the neck boost ($\times 2.41$ for the boosted bones after renormalisation) left it unchanged, and a geodesic loss moved neck02 only from 175.6 to 175.0 deg, likely because neck02 and head must turn back together. So I corrected the output with one constant rotation per flipped bone (Section 4.6): 178.8, 179.2, 179.8 and 179.7 deg for neck02, head, metacarpal1.L and finger4-1.R.

| | Base | corr., no train. | corr. + 2-ep. FT | 2-ep. control | Control 10 ep. | Final 10 ep. |
|---|---|---|---|---|---|---|
| neck02 (deg) | 175.6 | 4.5 | 4.7 | 175.7 | 175.8 | 4.8 |
| head bone (deg) | 177.9 | 12.4 | 11.2 | 177.9 | 177.7 | 8.9 |
| metacarpal1.L (deg) | 180.0 | 0.0 | 0.0 | – | 180.0 | 0.0 |
| finger4-1.R (deg) | n.d. | 10.2 | 10.3 | – | 180.0 | 9.6 |
| neck girth, pred./true | 0.461 | 1.004 | 0.997 | 0.461 | 0.464 | 0.996 |
| $PVE_{neck}$ (mm) | 27.3 | 10.8 | 10.7 | 27.9 | 27.7 | 11.0 |
| body PVE (mm) | 70.5 | 71.3 | 68.8 | 76.9 | 77.4 | 73.7 |
| PA-PVE (mm) | 53.3 | 52.2 | 50.9 | 55.2 | 56.4 | 50.3 |
| recall (%) | 90.2 | 90.2 | 89.4 | 91.1 | 89.0 | 87.0 |
| head direction (deg) | 12.0 | 19.1 | 14.4 | 13.8 | 13.0 | 13.1 |
| fingers (deg) | 17.3 | 8.2 | 8.3 | 17.4 | 17.4 | 8.3 |

**Table F.4.:** *The half-turned bones and their correction. V1, 100-image AnnyOne holdout, matched people. Bone errors: local medians (about 180 deg = half a turn). $PVE_{neck}$ over the 171 neck-skin vertices. Corr.: the bone correction (four constants from 200 training images); FT: fine-tune. 2-epoch columns: AnnyOne-only fine-tunes of Base; Control and Final: 10 epochs with 30% 3DPW. n.d.: not dumped. One run per column, no seed; PVE gaps below about 10 mm are within run-to-run variation (Table F.7).*

Without training the correction restores neck02, girth and neck-skin error, but the head direction worsens to 19.1 deg, above the 15-degree limit set beforehand, likely because the head's error differs per person; a 2-epoch fine-tune brings it to 14.4 deg (control 13.8). The small body-PVE differences need repeated runs to interpret. Figure D.2 shows two more holdout pairs. The original cause of the flips remains unresolved: the singular 6D initialization may explain the tilted metacarpal, but not the neck twist. The real-photo neck examples have no reference geometry and are judged visually (Figure 5.7).

## F.5. The fingers

In Table F.5 predicted fingers are 17.3 deg off and fingers at rest 15.1 deg: the prediction was worse than none. Hands are small (2.25 tokens long at the median, under two tokens for 41 %), but a person crop at test time leaves the finger error at 17.2 deg while body PVE nearly doubles, and a region-normalised rotation loss (a rebalance from about 70 to 33 % of the rotation loss, not an upweighting) changed nothing. The per-bone search on training images then found metacarpal1.L and finger4-1.R half-turned; correcting them roughly halves the finger error, with and without training.

About half the finger error was two flipped bones, not resolution. A hand-crop branch was never tried, and a rest-pose floor with the correction was not measured. Fingers on real photos: Figure 5.8.

| *(a) finger rotation error on the holdout* setting | fingers (deg) | reference (deg) |
|---|---|---|
| Base, predicted fingers | 17.3 | – |
| Base, fingers set to rest pose | 15.1 | – |
| Base, test-time person crop (PVE 124.8 mm) | 17.2 | 17.3 (paired) |
| region-normalised rotation loss, 2 ep. | 17.5 | 17.4 (control) |
| Base + bone correction, no training | 8.2 | 17.3 (Base) |
| Base + bone correction, 2-ep. fine-tune | 8.3 | 17.4 (control) |
| Final (10 ep., 30% 3DPW) | 8.3 | 17.4 (Control) |
| Fresh-heads-L, three metacarpals corrected | 17.8 | 32.0 (uncorr.) |

| *(b) hand size in tokens (14-px patches of the 672-px input)* | median length | under 2 tokens |
|---|---|---|
| full image, all hands | 2.25 | 41% |
| person crop at test time | 4.2 | 0% |
| children's hands, full image | 0.7 | 80% |

**Table F.5.:** *Fingers. (a) Finger rotation error, V1, 100-image AnnyOne holdout, matched people, against the matching run without the change; rest pose keeps the flipped metacarpal1.L. (b) Hand length in input tokens. One run per row, no seed.*

## F.6. Children and body shape

| *(a) predicted minus true phenotype, 10 holdout images (draft measurement)* phenotype | true mean | true std | weight 1 | weight 20 (Base) | reading |
|---|---|---|---|---|---|
| age | 0.553 | 0.036 | +0.199 | −0.002 | near the mode |
| height | 0.579 | 0.026 | −0.154 | −0.026 | near the mode |
| proportions | 0.763 | 0.070 | −0.017 | −0.001 | near the mode |
| firmness | 0.757 | 0.039 | −0.167 | −0.190 | unchanged |
| gender | 0.694 | – | −0.042 | −0.063 | ≈ |
| weight | 0.382 | – | +0.056 | +0.064 | ≈ |

| *(b) Base on the full holdout and in the young tail* | |
|---|---|
| holdout, all 222 people (nobody younger than 0.513) | age bias −0.015 (abs. 0.037); gender −0.008 (0.068) |
| holdout, true age 0.6–0.8 ($n$ = 53, mean 0.67) | predicted age 0.57 |
| 50 youngest training people (in-sample, true age 0.015) | predicted age 0.10; height −0.25 |
| training medians (AnnyOne) | gender 0.588, age 0.601, height 0.500 |

**Table F.6.:** *Shape head, phenotype units* $[0, 1]$*. (a) The stage before Base (shape weight 1) against Base (weight 20) on the same 10 holdout images. (b) Base, V1, on the full holdout and the 50 youngest training people (in-sample). In both tails predictions move toward the training median.*

Table F.6(a) is the draft's shape-weight test on 10 holdout images: raising the shape weight from 1 to 20 moved the age bias from +0.199 to −0.002 and the height bias from −0.154 to −0.026. Part (b) shows that this holds near the mode only: in both tails predictions move toward the training median (the

holdout has no children, so the young tail is in-sample). On Relative Human (Table 5.6) Base reads children as short adults: kids get a mean age of 0.61, babies 0.62 and adults 0.63 (AUC 0.61), none of 89 kids and 1 of 38 babies fall below the child threshold; the smallness goes into height instead. Gender carries no signal (accuracy 46.5 %). On our demo photos the medians of all 299 shape lines (gender 0.59, age 0.61, height 0.50) equal the training medians.

The child and gender results suggest a shared problem: on real photos, several shape predictions stay near the training median. Two mechanisms fit, and the evidence cannot rank them: the gap between synthetic and real images, or the loss (an L1 on sigmoid outputs regresses to the median, the sigmoid's gradient is 0.25 at 0.5 but 0.015 at 0.015, and only 2.53 % of training people are below age 0.3). It looks like a short adult because Anny's child shape (shoulder over head width 2.24 at age 0.1, 3.00 at 0.6) appears only below an age of about 0.4–0.5: a low height at age 0.6 gives a short adult (at least 1.22 m), and proportions barely matter (leg over stature −0.016).

These tests do not identify which mechanism dominates. I also avoid reporting age in years because the available age calibrations disagree.

## F.7. Pilot noise and restart cost

| Quantity | Observed values | Interpretation |
|---|---|---|
| Two identical controls: PVE | 76.3 vs. 68.7 mm | 7.6 mm difference between these runs |
| Head direction | 13.6 vs. 11.7 deg | 1.9 deg difference |
| Recall | 0.8 points apart | different matched sets |
| Six arms, Base recipe | PVE 68.7–76.3 mm; recall 88.6–90.2 %; head 11.3–13.7 deg | observed range, not a confidence interval |
| Restart comparison | 72.9 vs. 70.5 mm for Base | 2.4 mm higher mean PVE after restart |
| Root error | fix 0.53–0.73 m; controls 1.11–1.44 m | lower in every tested fix run |

**Table F.7.:** *Observed variation in the 2-epoch pilots, fine-tuned from Base at learning rate $10^{-5}$ without a fixed seed. V1, 100-image AnnyOne holdout, matched people. The restart comparison uses the unchanged-recipe controls. Two repeated pilots do not establish a general significance threshold or predict the variation of the final models.*

Table F.7 records the differences between two identical pilot configurations. These observations motivate matched controls and caution with small single-run differences; they are not a statistically estimated significance threshold for all experiments. The large changes in bone angles, neck girth and placement are supported by their targeted diagnostics. The PVE costs of the depth fix and neck pilot remain uncertain; no repeated final runs or bootstrap intervals were available.

## F.8. What did not work

Six changes did not fix a failure (Appendix E, Table E.1): an age-balanced sampler (inconclusive on 10 in-sample children), a global head orientation loss (within the spread of two identical controls), a region-normalised hand loss, a geodesic rotation loss, the neck and hand boost, and a $\times 20$ depth loss (no gain over $\times 5$). These tests did not provide reliable fixes under the tested settings. The placement and bone corrections were selected after the more specific diagnostics identified their target errors.

# G. Supporting Conversion Experiments

This appendix preserves the converter development and label-integrity checks that support the final procedure of Chapter 3. Historical versions are not the labels used for the main results.

## G.1. Development of the converter

The first converter (v1) chose the Anny bones by name. It gave 34.5 mm on `downtown_bar_00`; on four test scenes the error was 30 to 37 mm on every frame, and all four actors came out with the same body. This pattern suggested that shape fitting was compensating for a fixed mismatch between the skeletons. Placing both models in their rest poses showed two wrong choices that no name reveals. Anny's rig numbers the spine top-down, SMPL bottom-up, so SMPL's lumbar `spine1` had been mapped to a bone 25 cm above it; and SMPL's hips had been mapped to `pelvis.L/R`, 4 cm from the hip joint, instead of `upperleg01.L/R`. Version 2 picks the spine bones by height along the body axis and the hip bones by distance from the pelvis (Appendix B). Limbs and head fell to 1 to 10 mm, but the torso stayed at 43 to 73 mm (Table G.1).

The reason is that the rigs define their torso joints at different anatomical points: SMPL's pelvis joint sits 9 cm above its hips, Anny's root at hip height and 7 cm in front. For nine torso joints I therefore measure once, in the rest pose, the offset $\boldsymbol{\delta}_b$ from the Anny bone head to the SMPL joint in the bone's own frame, and the fit moves the *virtual joint* $\mathbf{p}_b = \mathbf{h}_b + \mathbf{R}_b\boldsymbol{\delta}_b$ onto the SMPL joint, with $\mathbf{h}_b$ and $\mathbf{R}_b$ the posed bone head and frame. The offsets are 11.6 cm (pelvis), 4.3 cm (hips), 3 to 7 cm (spine) and 5 cm (collars). With them (v2.1) the torso fell to 13 to 22 mm and four varied test scenes to 15.8 to 20.1 mm (one occluded frame 43.5 mm). Age, however, came out at exactly 0.67 for every actor in every version: joints cannot tell one adult age from another. Version 2.2 fixes age at 0.55; on all 87 person-sequences it fitted 22,915 frames with a median error of 18.2 mm. The converter's own error summaries did not reveal the label-integrity problems. They still held two of the three defects of the next section.

## G.2. Three label defects and the v2.3 re-fit

The converter grades its own fit, so it cannot see errors in what it is asked to fit or in what happens to the fits afterwards. I therefore wrote an independent

| converter | mean (mm) | max | worst joints (mm) |
|---|---|---|---|
| v1: correspondence by name, neutral SMPL targets | 34.3 | 37 | all joints, common floor |
| v2: correspondence by geometry, 3DPW joints, gender from metadata (reversed until v2.3) | 24.6 | 30 | spine1 73, collars 55–59, spine3 58, pelvis 53 |
| v2.1: + virtual torso joints, bone mask, warm start | 15.8 | 18.8 | shoulder 36, wrists 25–27 |

**Table G.1.:** *Mean joint error over all 24 joints on* `outdoors_fencing_01` *(38 frames, including full lunges) for three converter versions. The geometric correspondence is worth about 10 mm, the virtual joints a further 9 mm.*

check that reads the finished label files. On sampled images from every sequence it compares the labels with 3DPW's joints, surface and 2D keypoints, confirms that the loader passes them on unchanged, and tests gender, a label for every valid actor and the head turn. On the first label set ("old labels", v2.2 fits, about 2,000 images) its grade was FAIL, with three defects (Table 3.3).

**Defect 1: gender reversed.** The converter wrote 1.0 for men and 0.0 for women, MakeHuman's convention and the reverse of Anny's, so every 3DPW man was labelled with a female body. Refitting with the flipped value lowered the joint error for 72 of 73 men (median 19.2 to 15.3 mm); for women it raised the joint error (13.7 to 16.3 mm) but brought the surface closer for 27 of 31.

**Defect 2: missing actor.** In 407 of 960 sampled two-person images, one valid actor had no label. Each person's every-third-frame grid started at that person's own first valid frame, so after an invalid gap the two actors sat on different phases. The detection loss then treats a visible, real person as background.

**Defect 3: missing labels.** The training split had 4,451 label images where the fits implied 6,437 (31% missing). The image files of 8 sequences were incomplete on the cluster (`courtyard_captureSelfies_00`: 480 of 750 files), and the builder skipped frames without an image file without counting them. A rebuild after re-extracting the images recovered every label (Table 3.2, "rebuilt"), but still had defects 1 and 2, so I never trained on it.

**The re-fit.** Version 2.3 adds the two switches named above; their defaults reproduce v2.2 exactly. I re-fitted all 87 people with both switches on, and the post-check passed on all 60 sequences (Table 3.3, middle block). 9 single-person outdoor sequences became 1 to 3 mm worse, likely from a different warm start, which I did not test.

**The v23 labels.** I call the labels built from the v2.3 fits the v23 labels: 16,694 images and 22,749 people (Table 3.2). They have fewer images than the rebuilt

set but the same people, because both actors now share each frame. No person was dropped for fit error or box size. 163 frames were dropped for visibility, all from `outdoors_crosscountry_00`: with 3DPW's own camera, its ground-truth joints sit at a depth of −0.3 m, behind the camera. The 3D fit is fine; image labels for this sequence cannot exist with this camera, and no label set ever contained it.

**The check on the v23 labels.** The same check on the v23 labels (3,314 people in 2,360 images, 59 sequences) moved the grade from FAIL to WARN (Table 3.3). The median body-joint error fell from 18.7 to 14.1 mm, flipping the gender now makes the fit worse, and no sampled two-person image misses an actor. The head turn relative to the chest is now kept; in the old test labels it was mostly lost (6.3° against 20.0° in SMPL). Two rows stay at WARN, the limb-length ratios and the people with a joint more than 100 mm off, mostly at the hip, likely because Anny's and SMPL's limb proportions and hip joints differ (Table 3.3).

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